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BACHELOR THESIS

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Universal Deep Learning Detectors for Macromolecule Localization in Cryo-ET

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Abstract

This thesis studies the current practical status of *universal* deep learning detectors for macromolecule localization in cryo-electron tomography, with ProPicker as the central experimental platform. The work does not propose a new architecture from scratch. Instead, it develops a reproducible research repository, studies the current limits of ProPicker under realistic constraints and turns those observations into an optimized experimental pipeline: little supervision, domain shift between datasets and uncertainty about the robustness of the prompt itself.

The experimental program combines one real and one synthetic setting. First, ProPicker is evaluated on cytoplasmic ribosomes from EMPIAR-10988 as a real-data proof of concept. Then, the analysis moves to the UMU synthetic dataset with thyroglobulin, which provides full ground truth and controlled train/validation splits. On top of this baseline, the work studies incremental fine-tuning, compares single-prompt and multi-prompt inference and analyzes residual sensitivity to prompt orientation and prompt quality.

The results show a clear pattern. On EMPIAR-10988, prompt-only inference is informative but unbalanced, with an initial $F_1 = 0.3074$ that increases to 0.5714 after fine-tuning. On the UMU synthetic benchmark, the pretrained model collapses under domain shift with mean $F_1 \approx 0.008$, whereas fine-tuning raises performance to mean $F_1 = 0.891$. Incremental training reveals that most of the attainable performance is recovered early: using 12 training tomograms and multi-prompt inference yields the best observed configuration with $F_1 = 0.919$, while 4 tomograms already provide a highly competitive operating point. The final 125-prompt robustness study on `multi_inc16` shows that strong averages can still hide catastrophic prompt-level failures: the prompt-level mean is $F_1 = 0.780 \pm 0.283$, but several prompts collapse almost completely in recall. Weak descriptive trends appear for symmetry-axis orientation, embedding dispersion and prompt position inside the source tomogram, yet adjusted models and permutation tests do not support a single rotation-only explanation. The remaining bottleneck is prompt representativeness under anisotropic acquisition, not source SNR alone.

The main conclusion is that ProPicker is a promising basis for universal particle picking, but not yet a fully universal solution. Fine-tuning is essential under substantial domain shift, data efficiency is encouraging and the main unresolved bottleneck is prompt robustness: avoiding recall-collapse prompts whose failure is not explained by SNR or a simple angle-to-Z rule. The next logical step is a controlled multiclass benchmark built on the same synthetic split, together with explicit prompt-quality screening and symmetry-aware robustness controls.

Resumen extendido (*Extended Abstract*)

Contexto y motivación

La tomografía crioelectrónica permite observar macromoléculas en un contexto tridimensional cercano a su entorno nativo. Ese potencial es extraordinario desde el punto de vista biológico, pero llega acompañado de un problema computacional difícil: los tomogramas son grandes, ruidosos, anisotrópicos y muy heterogéneos. Antes de poder hacer subtomogram averaging, clasificación estructural o análisis espacial, hace falta localizar dónde están las partículas de interés. Esa etapa de *particle picking* sigue siendo uno de los cuellos de botella más importantes del flujo de trabajo.

En un escenario ideal, un detector moderno debería ser capaz de adaptarse a nuevas partículas y a nuevas condiciones de adquisición con muy pocos datos anotados. Esa idea es la que da sentido al concepto de *universal detector*. En vez de entrenar un modelo aislado para cada proteína y para cada dataset, interesa disponer de un sistema flexible que pueda transferirse con rapidez, que acepte alguna forma de condicionamiento mediante ejemplos o *prompts* y que permita refinarse con poco coste cuando el cambio de dominio lo exija. El trabajo desarrollado en esta memoria parte exactamente de esa pregunta: hasta qué punto un detector *promptable* como ProPicker puede comportarse de manera práctica y suficientemente general en cryo-ET real y sintético.

El proyecto no se planteó como el diseño de una arquitectura completamente nueva. El objetivo fue más concreto y más útil para el estado actual de la investigación: construir un entorno reproducible alrededor de ProPicker, documentar con rigor su comportamiento y responder una secuencia de preguntas experimentales que fuesen reduciendo la incertidumbre. Primero era necesario comprobar si el modelo preentrenado ofrecía señal útil en un caso real. Después había que medir qué ocurría cuando el dominio cambiaba de forma fuerte. A continuación resultaba imprescindible cuantificar cuántos tomogramas anotados hacían falta realmente para entrar en un régimen de rendimiento alto. Por último, una vez despejadas las dudas más gruesas sobre viabilidad y adaptación, quedaba por entender si la variabilidad restante dependía sobre todo del volumen analizado o de la calidad geométrica y representacional del *prompt*.

Objetivos y metodología

Con esa lógica se definieron cinco objetivos. El primero fue establecer un protocolo reproducible de preprocesado, ajuste fino e inferencia que pudiera mantenerse estable a lo largo de todos los experimentos. El segundo consistió en validar ProPicker sobre datos reales mediante una prueba de concepto con ribosomas en EMPIAR-10988. El tercero fue estudiar la adaptación a un dominio sintético controlado con ground truth completo en la colección UMU. El cuarto fue analizar eficiencia de datos y estrategias de *prompting*, en particular la diferencia entre usar un único embedding y usar un embedding agregado a partir de varias instancias. El quinto fue examinar si la robustez aparente del sistema escondía una dependencia fuerte de la orientación o de la calidad del *prompt*.

La implementación se organizó como un marco experimental reproducible con criterios estables de preprocesado, entrenamiento e inferencia. Cada experimento combina análisis exploratorio, selección de *prompts*, evaluación cuantitativa e inspección volumétrica, pero siempre dentro del mismo flujo metodológico: lectura de tomogramas en formato MRC, inversión de contraste cuando el modelo lo requiere, preparación de coordenadas, extracción de subtomogramas de tamaño $37 \times 37 \times 37$, generación de embeddings mediante el encoder de *prompts* de ProPicker y posterior inferencia o ajuste fino dentro de la misma cadena de procesamiento. Los mapas de localización generados por el modelo se convierten a coordenadas mediante un procedimiento de agrupamiento coherente a lo largo de toda la memoria. Cuando resulta útil para la inspección visual, esos mapas se exportan también a MRC para abrirlos en ParaView.

El conjunto de datos real utilizado fue EMPIAR-10988 en el caso `cyto_ribosome`. La configuración implementada en el repositorio usa `TS_029` para entrenamiento y `TS_030` para validación, con un recorte centrado definido por `crop_delta=64`. Esta decisión hace que el experimento sea una validación controlada y no un benchmark exhaustivo del dataset completo, pero precisamente por eso resulta útil como primera prueba de concepto. El conjunto sintético principal fue UMU synthetic data. En ese caso se fijó una partición estable de 25 tomogramas, con 20 para entrenamiento y 5 para validación. Esa partición se mantuvo constante desde el experimento 2 hasta el 4 para que las comparaciones entre curvas y configuraciones fuesen internamente válidas.

Las métricas principales fueron precisión, *recall* y F_1, junto con los conteos de verdaderos positivos, falsos positivos y falsos negativos. La memoria respeta las definiciones de evaluación adoptadas en el marco experimental. En el caso de EMPIAR-10988 se sigue el criterio de evaluación heredado del protocolo de referencia usado por los autores de ProPicker. En el caso UMU la comparación se realiza sobre coordenadas predichas frente a coordenadas de referencia en el conjunto de validación fijo. Aunque algunos umbrales concretos proceden de ese protocolo metodológico, todas las comparaciones del trabajo son coherentes porque dentro de cada benchmark se reutiliza exactamente el mismo procedimiento.

Desarrollo experimental

El primer experimento tuvo como finalidad comprobar si el modelo preentrenado era capaz de transferirse a un problema real antes de cualquier adaptación específica. La hipótesis era prudente: no se esperaba una solución terminada, pero sí una señal útil que justificase seguir invirtiendo trabajo en el pipeline. Eso fue exactamente lo que ocurrió. ProPicker base detectó ribosomas en EMPIAR-10988 con una precisión alta en relación con su *recall*, lo que evidenciaba una fuerte tendencia conservadora. En la primera evaluación obtuvo Precision = 0,7049, Recall = 0,1966 y F_1 = 0,3074, con 547 verdaderos positivos, 229 falsos positivos y 2236 falsos negativos. Tras optimizar el umbral de inferencia para maximizar F_1, el resultado cambió a Precision = 0,5125, Recall = 0,2648 y F_1 = 0,3492. El salto no resolvía el problema, pero demostraba que parte de la limitación estaba en la calibración y no en una ausencia total de información útil.

La segunda fase del experimento 1 aplicó fine-tuning sobre el volumen recortado con las anotaciones del caso ribosomal. Ahí se produjo el cambio cualitativo relevante. El modelo afinado alcanzó Precision = 0,5677, Recall = 0,5752 y F_1 = 0,5714, con 1044 verdaderos positivos, 795 falsos positivos y 771 falsos negativos. La lectura correcta de este resultado no es que el fine-tuning produzca una mejora incremental, sino que cambia de régimen al detector. El *recall*, que en el modo *out-of-the-box* era claramente insuficiente, deja de ser el cuello de botella principal. Con esto quedó validada la primera tesis operativa del proyecto: el paradigma *prompt-based* es útil para arrancar, pero cuando se busca rendimiento equilibrado sobre datos reales el ajuste fino deja de ser opcional.

El segundo experimento llevó el análisis al dominio sintético UMU con partícula `thyroglobulin`. Aquí la pregunta era más dura, porque ya no se trataba de mejorar un comportamiento prometedor, sino de medir un cambio de dominio severo con ground truth completo. Los resultados fueron muy claros. El modelo base prácticamente colapsó en este contexto: la media final fue Precision $\approx$ 0,009, Recall $\approx$ 0,008 y $F_1 \approx$ 0,008. En otras palabras, fuera de su dominio de preentrenamiento y con una partícula más compleja, la inferencia basada solo en *prompt* no era operativa. Sin embargo, cuando se aplicó fine-tuning sobre los 20 tomogramas de entrenamiento fijados, la situación cambió por completo. En el conjunto de validación formado por `tomo_rec_5` a `tomo_rec_9`, la media fue Precision = 0,822, Recall = 0,983 y F_1 = 0,891.

Ese resultado fue además bastante homogéneo salvo en el caso más ruidoso del conjunto. Los tomogramas `tomo_rec_5_snr1.66`, `tomo_rec_6_snr1.17`, `tomo_rec_7_snr1.13` y `tomo_rec_9_snr1.28` se movieron en una banda F_1 entre 0.923 y 0.944. El caso `tomo_rec_8_snr0.57` cayó a F_1 = 0,706 debido a una precisión de 0.565, mientras que el *recall* se mantuvo en 0.939. Ese patrón fue importante porque señaló desde muy pronto dónde residía la debilidad restante: una vez corregido el cambio de dominio, el problema principal ya no era dejar partículas sin detectar, sino controlar falsos positivos cuando el volumen era más difícil.

El tercer experimento respondió a una cuestión práctica decisiva: cuánta anotación hace falta realmente para obtener buen rendimiento y si un esquema *multi-prompt* compensa frente al uso de un único embedding. Para ello se reutilizó exactamente el mismo conjunto de validación del experimento 2 y se entrenaron incrementos de 1, 2, 4, 8, 12, 16 y 20 tomogramas, con un número de épocas adaptado a cada tamaño muestral. El resultado más útil fue que la curva de aprendizaje sube deprisa y luego entra en meseta. En `single_prompt` los F_1 medios fueron 0.691, 0.728, 0.800, 0.860, 0.862, 0.888 y 0.849. En `multi_prompt_n10` fueron 0.647, 0.717, 0.875, 0.906, 0.919, 0.911 y 0.894.

La interpretación de esas curvas es muy rica. Primero, con solo 4 tomogramas ya se alcanza aproximadamente el 93.1 % del mejor resultado observado, lo que convierte esa configuración en un punto de alta eficiencia de datos. Segundo, la mejor configuración concreta del estudio fue `multi_prompt_n10` con 12 tomogramas y F_1 = 0,919. Tercero, la ventaja del *multi-prompt* no aparece sobre todo por incrementar el *recall*, sino por reducir falsos positivos. En el incremento 12, el modo `single_prompt` obtuvo Precision = 0,773, Recall = 0,982, F_1 = 0,862, 611 verdaderos positivos, 190 falsos positivos y 11 falsos negativos. El modo `multi_prompt_n10` produjo Precision = 0,894, Recall = 0,947, F_1 = 0,919, 590 verdaderos positivos, 70 falsos positivos y 32 falsos negativos. El balance es muy favorable al esquema multiinstancia porque reduce de forma fuerte el ruido de decisión con una pérdida moderada de sensibilidad.

El cuarto experimento se centró en una pregunta más fina: si el modelo ya rinde bien de media después del fine-tuning, por qué algunos *prompts* siguen funcionando mucho mejor que otros. La versión final del estudio amplió de forma sustancial el tamaño muestral y reformuló la hipótesis. En lugar de limitarse a 10 *prompts* rotacionalmente diversos, el análisis definitivo generó una población de 200 candidatos a partir de los tomogramas de entrenamiento y estudió los primeros 125 sobre el checkpoint `multi_inc16`. El objetivo ya no era solo preguntar si existe una «mala rotación», sino separar con más cuidado tres explicaciones posibles: calidad del tomograma de origen, anisotropía de adquisición por *missing wedge* y ambigüedad geométrica del *prompt* en una partícula con simetría C2 como thyroglobulin.

El resultado principal fue doble. Por un lado, el rendimiento medio siguió siendo bueno: $F_1 = 0{,}780 \pm 0{,}283$, precisión media 0.824 y *recall* medio 0.783 sobre los 125 *prompts*. Por otro, esa media escondía fallos catastróficos. Ocho *prompts* obtuvieron *recall* medio igual a cero, y el *prompt* 124 alcanzó precisión media 1.000 con *recall* medio 0.115. Esa combinación es muy informativa: el problema dominante ya no es una avalancha de falsos positivos, sino una activación insuficiente del detector. En términos prácticos, un *prompt* malo es aquel que deja de representar bien a la clase y provoca un colapso del *recall* en tomogramas no vistos.

La interpretación también cambió de forma importante. El SNR del tomograma de origen siguió siendo una explicación débil ($r \approx +0{,}05$ con el *recall* medio), y los descriptores orientacionales simples tampoco fueron suficientemente fuertes por sí solos. Sí aparecieron tendencias descriptivas compatibles con la teoría física: *prompts* cuyo eje C2 se aleja del eje global Z tienden a mejorar ligeramente, y las posiciones dentro del tomograma importan más de lo esperado, con correlaciones más visibles para la distancia al centro y la distancia a los bordes que para el SNR de origen. Sin embargo, cuando se ajusta un modelo causal con controles de calidad, posición y efectos fijos por tomograma de origen, el bloque rotación+adquisición solo añade $\Delta R^2 = 0{,}0109$ para F_1 y $\Delta R^2 = 0{,}0252$ para *recall*, con tests conjuntos y permutacionales claramente no significativos. En otras palabras, la hipótesis «los *prompts* malos son sobre todo malas orientaciones» no queda respaldada como explicación dominante.

La lectura final es más precisa y, a la vez, más útil. El problema residual de ProPicker no es simplemente la rotación ni simplemente el ruido. Es la representatividad del *prompt* bajo una cadena completa de efectos: simetría C2, adquisición anisotrópica, contexto local y centrado del subtomograma. Por eso, al cierre de la memoria, la barrera principal ya no parece ser la cantidad de datos anotados, sino la capacidad de detectar y evitar *prompts* no representativos antes de la inferencia.

Conclusiones y estado actual de la investigación

Visto en conjunto, el programa experimental deja una imagen bastante nítida del estado actual. La viabilidad básica del enfoque está demostrada. ProPicker no parte de cero en datos reales y, tras fine-tuning, alcanza niveles de rendimiento altos y muy competitivos en el dominio sintético controlado. La eficiencia de datos es mejor de lo que cabría esperar en un problema tan complejo. Con pocos tomogramas bien elegidos ya se captura una fracción muy alta del rendimiento máximo. Además, el esquema multiinstancia aporta una mejora práctica clara al reducir falsos positivos.

Al mismo tiempo, la memoria no esconde las limitaciones que siguen abiertas. La universalidad todavía es condicional. En presencia de cambios de dominio intensos el modelo preentrenado necesita adaptación. Los experimentos completados se apoyan en un único benchmark real y en un benchmark sintético monoclase. La invariancia al *prompt* tampoco puede darse por resuelta: hay una fracción del comportamiento del sistema que depende de qué partícula concreta se use como referencia y de cómo esa referencia se proyecta en el espacio de embeddings. Por eso el proyecto ha pasado de una fase de validación gruesa a una fase de refinamiento fino.

El estado del repositorio refleja exactamente esa transición. A marzo de 2026 están implementados y documentados los experimentos 1 a 4, junto con la infraestructura de entrenamiento, inferencia, exportación y análisis. También está redactada la especificación del experimento 5, centrado en fine-tuning multiclase sobre tres clases de UMU synthetic: `thyroglobulin` como continuidad natural del trabajo anterior y dos clases adicionales, `pdb_4v94` y `pdb_4cr2`, elegidas por ofrecer un equilibrio razonable entre ambición y control. Ese experimento no se ha ejecutado todavía y por tanto forma parte del trabajo futuro, no de los resultados cerrados de esta memoria.

La conclusión central es, por tanto, doble. Por un lado, el enfoque *promptable* apoyado por fine-tuning sí constituye una base sólida para construir detectores cada vez más generales en cryo-ET. Por otro, el adjetivo «universal» todavía debe entenderse como una meta de investigación y no como una propiedad ya alcanzada. El sistema actual ha resuelto la duda de viabilidad. La duda que queda abierta es una duda de robustez fina, de extensión multiclase y de transferencia más amplia a datos reales. Precisamente por eso los siguientes pasos están bien definidos y el proyecto entra en una fase especialmente fértil.

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Terms and Acronyms

Cryo-ET Cryo-electron tomography.

EMPIAR Electron Microscopy Public Image Archive.

UMU University of Murcia.

Tomogram 3D volume reconstructed from a tilt series.

Tilt series Set of projection images acquired at different tilt angles.

Missing wedge Region of unsampled Fourier space caused by the limited tilt range.

Subtomogram Small 3D crop extracted around a candidate particle location.

STA Subtomogram averaging.

Particle picking Identification of particle coordinates in a tomogram.

Prompt A particle-specific reference crop or embedding used to condition ProPicker.

ProPicker Promptable 3D segmentation model for particle picking in cryo-ET.

TomoTwin Encoder used in the repository to generate prompt embeddings.

DeepETPicker Preprocessing and training framework reused by the ProPicker toolchain.

CNN Convolutional neural network.

U-Net Encoder-decoder CNN with skip connections for dense prediction.

FiLM Feature-wise linear modulation.

SNR Signal-to-noise ratio.

IoU Intersection over union.

NMS Non-maximum suppression.

PCA Principal component analysis.

t-SNE t-distributed stochastic neighbor embedding.

FFT Fast Fourier transform.

MRC Electron microscopy volume file format commonly used for tomograms and density maps.

SO(3) 3D rotation group.

C_2 Two-fold rotational symmetry.

F_1 Harmonic mean of precision and recall.

Chapter 1

Introduction

1.1. Motivation

Cryo-electron tomography has become one of the most powerful techniques for observing macromolecular organization *in situ*. By reconstructing a 3D tomogram from a tilt series of a vitrified specimen, cryo-ET preserves spatial context that is usually lost in purified single-particle pipelines. This advantage is especially relevant when the scientific question concerns intracellular organization, molecular crowding or the coexistence of several complexes within the same environment [7, 1].

The computational burden begins immediately after reconstruction. Tomograms are noisy, anisotropic and large, and several downstream stages depend on knowing where the particles of interest are located. Subtomogram extraction, alignment and averaging require coordinate estimates that are accurate enough to seed the rest of the structural analysis workflow [2, 11]. For this reason, particle picking is one of the main gates that determine whether a cryo-ET study scales or stalls.

Manual annotation does not scale to current datasets, while classical methods such as template matching often require strong assumptions about the target structure and can become brittle when the particle changes conformation, size or local environment. These constraints motivate the search for more flexible detectors that can retain good performance under heterogeneous acquisition conditions and under low-label regimes.

This thesis is framed around that goal. It studies the current state of *universal* deep learning detectors for cryo-ET localization by focusing on ProPicker, a promptable segmentation model that can either work from a prompt alone or be fine-tuned to a target particle with limited data [14]. The work is not purely algorithmic. Its main value lies in establishing a reproducible experimental workflow and in identifying, with quantitative evidence, where the current generation of promptable detectors is already useful and where it still breaks down.

1.2. Problem statement

The problem addressed in this thesis is the localization of macromolecules in cryo-ET tomograms with a detector that aims to be as flexible as possible across particles and datasets.

Given a tomographic volume $V \in \mathbb{R}^{X \times Y \times Z}$, the target is to estimate a set of detections

$$\mathcal{D} = \{(\mathbf{p}_i, s_i, c_i)\}_{i=1}^{N},$$

where $\mathbf{p}_i \in \mathbb{R}^3$ denotes the predicted particle center, s_i is a confidence-related score or response derived from the localization map and c_i is the particle identity in single-class or future multiclass settings.

In practice, the problem is difficult because cryo-ET data combine several adverse conditions:

- very low SNR caused by dose limitations;
- anisotropic resolution due to the missing wedge;
- crowded cellular or synthetic scenes with structured background;
- limited annotations and strong cost of manual labeling;
- computational constraints imposed by large 3D volumes.

The specific focus of this thesis is narrower than the full detection problem. The question is not whether a detector can achieve good performance on a single curated benchmark under extensive supervision, but whether a *promptable* detector can remain useful when supervision is scarce, the domain changes and prompt quality is itself a variable under study.

1.3. Research questions

The work is organized around the following research questions:

1. Can prompt-only inference with ProPicker provide useful transfer on real cryo-ET data before any task-specific fine-tuning?
2. How severe is the performance drop when the detector is moved to a substantially different domain?
3. How much annotated data is required to recover strong performance after fine-tuning?
4. Does aggregating several prompts improve practical robustness with respect to a single prompt?
5. Once fine-tuning is effective, is the remaining variability driven mainly by tomogram difficulty or by prompt-dependent factors such as orientation and embedding quality?

1.4. Contributions

The main contributions of the thesis are:

1. A reproducible research repository that organizes preprocessing, prompt construction, fine-tuning, inference and result export for ProPicker-based cryo-ET particle picking.
2. A real-data proof of concept on EMPIAR-10988 showing that prompt-only inference is viable but insufficient for balanced performance.
3. A controlled synthetic benchmark on UMU thyroglobulin demonstrating that fine-tuning is essential under severe domain shift.

4. A data-efficiency study that identifies a strong operating point with only 4 training tomograms and the best observed configuration at 12 tomograms with multi-prompt inference.
5. A prompt-robustness analysis showing that residual instability is expressed mainly as prompt-dependent recall collapse and is only weakly explained by source SNR or simple rotation proxies taken in isolation.
6. An evidence-based optimized pipeline that specifies how to preprocess data, choose prompts, fine-tune and audit failures when ProPicker is deployed on a new benchmark.

1.5. Document structure

Chapter 2 reviews the cryo-ET localization problem and the background needed to understand promptable detectors. Chapter 3 describes the objectives, datasets, methodology and evaluation protocol. Chapter 4 presents the implementation and the experimental development carried out in the repository. Chapter 5 reports the results of the four completed experiments and discusses their implications. Chapter 6 summarizes the conclusions and outlines the future work. The appendix documents the hardware and software environment used in the project.

Chapter 2

State of the Art

2.1. Cryo-electron tomography fundamentals

Cryo-ET reconstructs a 3D volume from a series of projection images acquired at different tilt angles. Because the angular range is limited in practice, Fourier space is sampled incompletely and the reconstruction exhibits the well-known missing wedge artifact [7, 1]. This causes direction-dependent distortions and makes pattern recognition intrinsically harder than in isotropic 3D imaging settings.

At a workflow level, the acquisition and reconstruction process can be summarized in four steps. First, a tilt series of 2D projections is acquired while the specimen is mechanically tilted. Second, the images are motion-corrected and aligned, usually with fiducials or alignment models that compensate for grid deformation. Third, a 3D tomogram is reconstructed, typically through weighted or filtered backprojection or through iterative methods such as SIRT and SART. Fourth, the reconstructed volume is used for downstream tasks such as particle picking, segmentation or subtomogram averaging [8, 17]. This sequence matters because the localization problem studied in this thesis is not applied to raw particle images but to reconstructed volumes that already contain the acquisition footprint of the microscope and the reconstruction algorithm.

From a machine learning perspective, cryo-ET data combine several characteristics that complicate detector training:

- signal is weak and often varies locally across the volume;
- structures can be small relative to the tomogram size;
- particles appear under different orientations and in different contextual neighborhoods;
- class labels are expensive to obtain and often noisy;
- preprocessing choices such as denoising, contrast normalization or reconstruction settings modify the data distribution seen by the detector.

Tools such as Warp have been instrumental in automating several preprocessing stages, including alignment-related steps and real-time handling of cryo-EM data streams [12]. Nevertheless, the output of these pipelines still needs specialized localization methods that remain stable under substantial variability.

Figure 2.1 gives a representative example of the type of volumetric reconstruction discussed in this chapter: cellular context, membrane organization and particle-scale structure are all present simultaneously in the same cryo-ET volume.

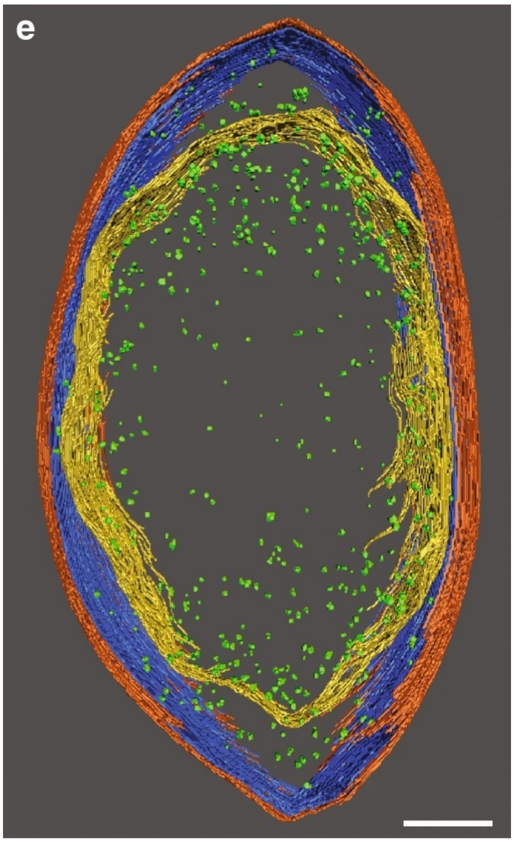

Figure 2.1: Example of a cryo-electron tomography 3D reconstruction showing membrane layers and ribosome locations inside a reconstructed cell.

Source: Taiki Katayama et al., Wikimedia Commons.

2.1.1. From tilt-series acquisition to rotation-sensitive prompts

The rotational-sensitivity problem studied later in EXP4 is not only a model-side artifact. It starts at acquisition. In an ideal isotropic 3D imaging system, rotating a particle would mostly produce a rotated copy of the same signal. Cryo-ET breaks that symmetry because the specimen is scanned over a limited angular range and around a single mechanical tilt axis. The resulting missing wedge removes direction-dependent Fourier information and creates anisotropic blur and elongation in the reconstructed tomogram [7, 1]. As a consequence, two subtomograms containing the same macromolecule at different orientations are not related by a clean rigid rotation alone.

This can be written schematically as

$$y \approx \mathcal{A}_{\mathrm{mw}}(Rx) + \eta,$$

where x is the underlying particle, R is its rigid orientation, η is noise and $\mathcal{A}_{\mathrm{mw}}$ is the acquisition-reconstruction operator induced by limited-angle tomography. The key point is that $\mathcal{A}_{\mathrm{mw}}$ is not rotationally neutral. Rotating the object and then reconstructing it is not equivalent to reconstructing it once and rigidly rotating the resulting volume. This is precisely why missing-wedge compensation methods such as IsoNet exist: the anisotropy induced by limited-angle acquisition degrades both visual interpretability and isotropy of the reconstructed signal [6].

This matters directly for promptable detection. A prompt crop does not contain only the target particle. It also contains the acquisition footprint left by the missing wedge, the local reconstruction context, possible centering error and whatever contrast distortions survive preprocessing. Therefore, a prompt that is geometrically favorable with respect to the acquisition axis can yield a much more stable representation than another prompt from the same class but at a less favorable orientation. In practical terms, prompt identity and prompt orientation become plausible explanatory variables even before any model-specific effect is considered.

The model architecture adds a second source of difficulty. Standard convolutional networks are naturally equivariant to translations, but not to arbitrary rotations [3]. In 3D this issue is more severe, because the relevant variability belongs to volumetric rigid motions and because enforcing rotation equivariance requires specialized constructions such as steerable 3D convolutions [15]. ProPicker gains flexibility by conditioning the detector on a prompt, but that conditioning mechanism does not make the overall pipeline automatically rotation-invariant [14]. This is why EXP4 should be interpreted as a targeted test of the full acquisition–reconstruction–prompt–model chain, not as a cosmetic ablation.

The specific target used in the synthetic benchmark adds a second, independent source of rotational ambiguity. Human thyroglobulin is a homodimer with global C2 symmetry, and the reported cryo-EM structure is approximately $250 \times 160 \times 110$ Å, with the shortest dimension aligned with the C2 axis [18]. Two poses that differ by a 180° rotation around that symmetry axis are physically equivalent. For a pose-sensitive learning system, the natural space is therefore not pure SO(3), but $SO(3)/C_2$. Any formulation that ignores this symmetry implicitly asks the model to separate states that the structure itself does not distinguish.

For thyroglobulin, the problematic rotations are therefore of two kinds. The first are symmetry-equivalent rotations, which should ideally be collapsed by the loss or by the evaluation metric. The second are acquisition-amplified ambiguities: views for which the discriminative differences between poses lie mostly along the anisotropic direction degraded by the missing wedge. These two mechanisms interact. Near-symmetric views are already difficult to distinguish in $SO(3)/C_2$, and missing-wedge anisotropy can make them even more degenerate after reconstruction. From a modeling perspective, this is why physically plausible 3D augmentations, symmetry-aware losses and training under matched missing-wedge conditions are attractive strategies in cryo-ET [15, 6, 17].

2.2. Macromolecule localization and subtomogram analysis

Particle picking seeks to estimate the coordinates of particle centers inside the tomogram. Those coordinates are then used to extract subtomograms, which can be aligned, classified and averaged in order to recover a higher-SNR structural signal [2, 11]. A weak picker damages the entire downstream workflow: if recall is low, particles are missed; if precision is low, later stages waste time on false positives or degrade due to contamination.

Traditional cryo-ET localization methods fall roughly into two groups. Template matching compares the tomogram with a reference template across positions and orientations.

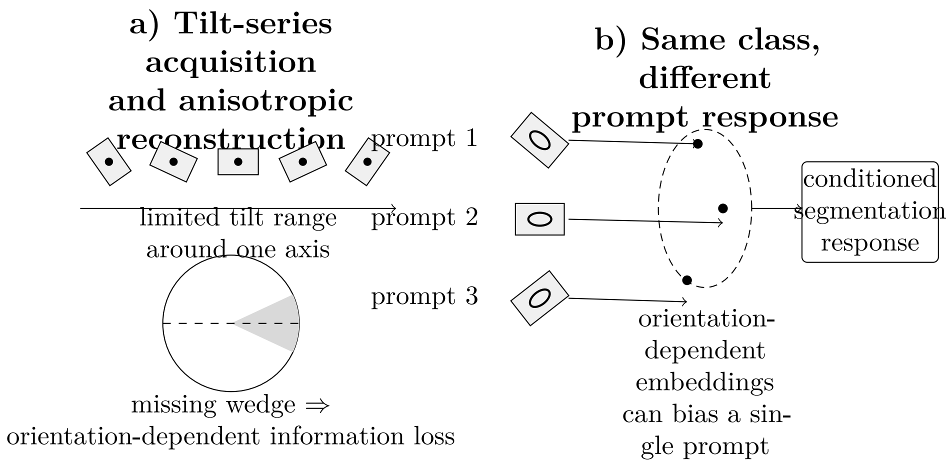

Figure 2.2: Conceptual schematic of why rotational invariance is difficult in cryo-ET. Limited-angle tilt acquisition produces anisotropic reconstructions with a missing wedge, so rotated particles are not observed as simple rigidly rotated copies after reconstruction. When such subtomograms are used as prompts, differences in orientation, reconstruction artifact and local context can lead to different embeddings and different detector responses. This is the conceptual motivation for the prompt-robustness study in EXP4.

Feature-based methods rely on filters, local maxima and hand-designed rules. Template matching remains relevant in some settings, but it can be computationally expensive and it assumes that a sufficiently representative template is available. Hand-crafted strategies are usually more brittle when the target varies in appearance or when background clutter changes substantially.

2.3. Deep learning formulations for 3D particle picking

Modern learning-based approaches usually reframe particle picking as one of the following problems:

- voxel-wise segmentation or center heatmap prediction;
- 3D object detection on patches or subvolumes;
- similarity-driven retrieval and matching in embedding space.

Architectures inspired by U-Net remain common because they combine local detail with contextual information and can be adapted to volumetric inputs [10]. More general object detection literature distinguishes anchor-based and anchor-free formulations [9, 5, 13, 16]. In cryo-ET, however, the representation of interest is often a particle center rather than a tight 3D bounding box, so segmentation maps and center-localization maps are especially natural.

Figure 2.3 is included as a compact reminder of the encoder-decoder pattern that motivates many later 3D cryo-ET variants. In the present domain, the same skip-connection logic is typically retained while the 2D operators are replaced with volumetric blocks.

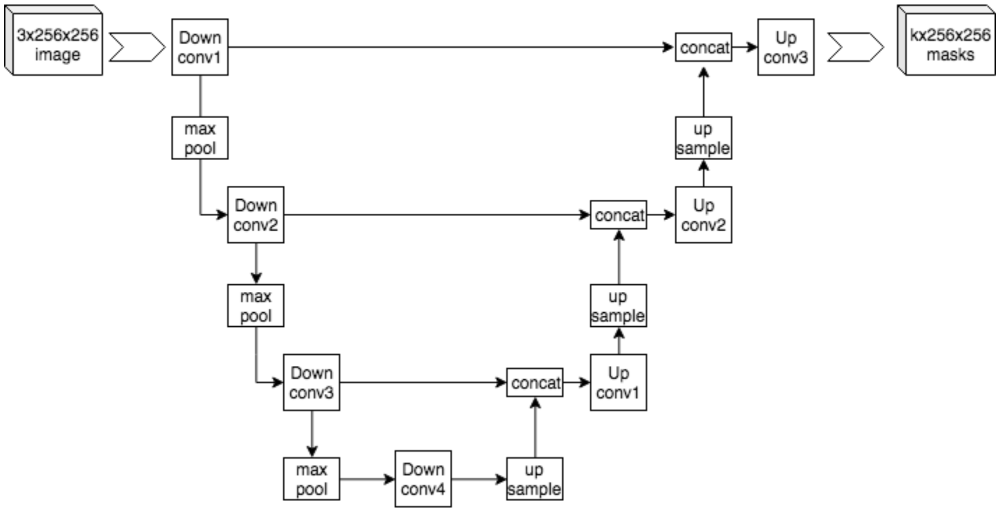

Figure 2.3: Illustrative U-Net encoder-decoder architecture. Cryo-ET implementations usually preserve this skip-connected design while replacing the 2D image operators with their 3D counterparts.

Despite the progress of generic 2D detectors [4], cryo-ET retains a clear volumetric character. Working with 3D patches is therefore advantageous, but it also imposes memory constraints that force careful tiling, padding and response aggregation.

2.4. Promptable detectors and the idea of universality

The notion of a universal detector in cryo-ET is closely tied to flexibility. A model that performs very well on one fixed class but requires retraining from scratch for every new particle is useful, but not universal in any practical sense. Promptable detectors attempt to bridge that gap by conditioning a general model on a reference example of the target particle.

ProPicker is one of the most recent examples of this family. It combines a prompt encoder with a 3D segmentation network that is conditioned on prompt-derived features and then produces a localization map from which coordinates can be extracted [14]. The main attraction of this design is twofold. First, prompt-only inference enables flexible targeting of new particles without retraining. Second, if that transfer is insufficient, fine-tuning can specialize the same model with comparatively little data.

This design is conceptually aligned with broader trends in promptable vision systems, but cryo-ET adds constraints that make the problem harder. A prompt is not always equally informative. Its orientation, local context, centering quality and embedding statistics may all affect the downstream response of the segmentation model. Therefore, universality is not only about training data diversity or network capacity; it is also about the stability of the conditioning mechanism.

2.5. Real and synthetic data in cryo-ET benchmarking

Real and synthetic tomograms serve complementary purposes. Real datasets capture the complexity of laboratory conditions, reconstruction artifacts and biological variation. Synthetic datasets, by contrast, provide full ground truth and allow controlled experiments on noise, density, class distribution and missing-wedge-related distortions. A serious evaluation program benefits from both.

That is precisely why this thesis combines EMPIAR-10988 and UMU synthetic data. The real dataset answers whether the detector is useful in a realistic scenario. The synthetic benchmark answers more controlled questions about adaptation, data efficiency and prompt robustness, because it permits stable train/validation splits and exhaustive metric computation.

2.6. Evaluation considerations

In localization tasks, performance is normally summarized through precision, recall and F_1, with matches defined by a geometric criterion between predicted and ground-truth particle coordinates. When the detector produces a dense response map, clustering and thresholding become part of the effective evaluation pipeline. This is especially important for cryo-ET because changes in the segmentation threshold can shift the balance between false positives and false negatives considerably.

For this reason, a key principle in this thesis is internal consistency. Within each benchmark, the same metric definition and the same post-processing rules are reused across all competing settings. This makes the comparisons trustworthy even when some low-level thresholds are inherited from the original methodological protocol and are not independently re-derived at every exploratory stage.

Chapter 3

Objectives and Methodology

3.1. Objectives

The thesis investigates the practical value of ProPicker as a basis for universal macromolecule localization in cryo-ET. Rather than optimizing a single leaderboard score, the explicit project objective is to study the current limits of ProPicker and to convert that characterization into a concrete, optimized pipeline for experimentation and later deployment.

The specific objectives are:

1. to build a reproducible repository that can support repeated fine-tuning and inference studies across several datasets;
2. to validate the complete workflow on real cryo-ET data;
3. to quantify domain-shift effects on a synthetic benchmark with full ground truth;
4. to measure data efficiency and compare prompt aggregation strategies;
5. to study the robustness limits of the current prompt-conditioning mechanism;
6. to leave the project in a state that naturally supports the next multiclass step.

3.2. Datasets

3.2.1. Experimental data: EMPIAR-10988

The real-data benchmark uses the EMPIAR-10988 ribosome case already supported by the ProPicker examples. In the repository, the experiment focuses on the `cyto_ribosome` target. Training is performed on tomogram `TS_029` and validation on `TS_030`. The workflow uses a centered crop with `crop_delta=64`, which yields a controlled subvolume of $128 \times 128 \times 128$ voxels around the region of interest.

This setup must be interpreted correctly. It is a proof of concept on real data, not a full-scale benchmark over the entire EMPIAR collection. That limitation is deliberate, because the first goal was to verify the pipeline from prompt extraction to fine-tuned inference under realistic data conditions while keeping the experiment small enough to debug carefully.

Because EXP1 focuses on cytoplasmic ribosomes, Figure 3.1 is included as a structural reference for the target macromolecule. It is illustrative rather than dataset-derived, but it helps anchor the discussion around the particle class being localized in the tomograms.

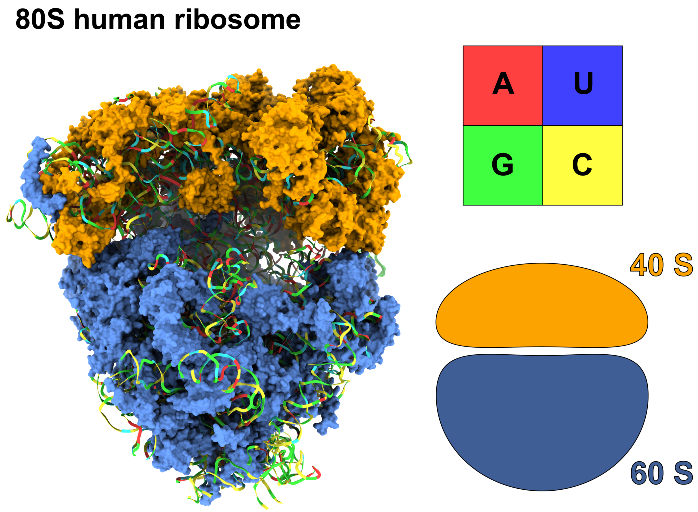

Figure 3.1: Illustrative 80S ribosome structural model used as a visual reference for the ribosome target discussed in the EMPIAR-10988 experiment.

Source: Ykust, Wikimedia Commons, CC BY 4.0.

3.2.2. Synthetic data: UMU benchmark

The main synthetic benchmark uses UMU synthetic tomograms. The completed experiments focus on the `thyroglobulin` class, corresponding to label 7 in the CSV annotations. The repository-level reports document 25 tomograms and approximately 3327 labeled thyroglobulin particles in total. The split used from Experiment 2 onward is fixed:

- 20 tomograms for training;
- 5 tomograms for validation: `tomo_rec_5_snr1.66`, `tomo_rec_6_snr1.17`, `tomo_rec_7_snr1.13`, `tomo_rec_8_snr0.57` and `tomo_rec_9_snr1.28`.

Keeping the validation set fixed is essential because Experiments 2, 3 and 4 are intended to be directly comparable.

3.2.3. Preprocessing and annotation strategy

The preprocessing workflow is deliberately conservative because the thesis aims to compare experimental factors rather than to re-optimize the entire toolchain at every exploratory stage. The shared pipeline can be summarized as follows:

1. input tomograms are read in MRC format and standardized into a common experimental layout;
2. contrast is inverted when needed so that the synthetic and real crops follow the ProPicker/TomoTwin convention of bright particles on dark background;

3. particle coordinates are expressed in the coordinate convention required by the detector and the evaluator;
4. prompt subtomograms of $37 \times 37 \times 37$ voxels are extracted around labeled particles;
5. prompt embeddings are generated with the TomoTwin encoder bundled with the repository workflow;
6. the preprocessing stage produces normalized tomograms, labels and train/test definitions that are reused consistently in fine-tuning and inference.

For UMU, the pipeline contains two additional transformations that are scientifically important. First, the annotation metadata are reconciled with the local identity of each tomogram so that the benchmark remains internally consistent. Second, the thyroglobulin coordinates are converted from Å to voxel units using the voxel-size metadata read from the tomograms. This prevents one of the most common reproducibility errors in synthetic cryo-ET pipelines, namely comparing predictions in voxel space against annotations that still live in physical coordinates.

3.3. Method overview

3.3.1. Adopted detector and design rationale

The detector used throughout the thesis is ProPicker [14]. The choice is motivated by its promptable design and by its ability to support two complementary modes:

- **prompt-only inference**, which tests how much cross-domain transfer can be obtained without task-specific training;
- **particle-specific fine-tuning**, which tests how efficiently the same model can adapt when transfer alone is insufficient.

This makes ProPicker a suitable platform for studying universality as a continuum rather than as a binary property.

3.3.2. Prompt construction and inference

Each prompt is built from a subtomogram centered on a labeled particle. The prompt encoder transforms the crop into an embedding that conditions the segmentation network. Inference then produces a localization map over the target tomogram, after which clustering and thresholding convert the response into discrete coordinates.

The thesis studies three prompt regimes:

1. one fixed prompt per experiment, used as the simplest baseline;
2. a multi-prompt embedding obtained from several instances, used to stabilize inference;
3. rotation-diverse prompt sets, used to probe orientation sensitivity.

The implementation became progressively more selective as the investigation advanced. In EXP1 and EXP2 the prompt is a fixed, manually validated positive instance, because the objective there is to measure transfer and domain adaptation, not prompt search. In

EXP3 the study introduces a robust prompt-ranking function in order to avoid a naive strategy based only on local variance. The physical-quality score used in that selector is

$$q_i = w_f \log(f_i) + w_c \log(c_i) - w_{dc} d_i,$$

where f_i is a mid-frequency to high-frequency Fourier power ratio, c_i is a center-to-border intensity ratio and d_i is a DC-penalty term that penalizes offset-dominated crops. The default weights are $w_f = 1.0$, $w_c = 0.9$ and $w_{dc} = 0.6$. The selector first ranks physically plausible candidates, keeps the best $k = 8$, and can in principle be extended with an embedding-consensus term. In the actual EXP3 single-prompt workflow, the final prompt was selected from the physical score alone.

The multi-prompt branch of EXP3 follows a different logic. It extracts ten candidate subtomograms, computes their individual embeddings with TomoTwin and represents the particle class in two ways: through the first embedding as a single-instance prompt and through the arithmetic mean of all ten embeddings as a multi-instance prompt. The selection stage does not simply take the ten most variant crops. Instead, it sorts candidates by variance and samples uniformly across that ranked list, which preserves diversity while avoiding a collapse onto extreme outliers.

EXP4 adds a third layer of control by selecting prompts in quaternion space. After quality filtering, prompts are chosen greedily to maximize the minimum angular separation

$$\theta(q_i, q_j) = 2 \arccos\left(|q_i \cdot q_j|\right),$$

with a target minimum distance of 30°. This is the operational definition of rotational diversity used throughout the robustness study.

3.3.3. Fine-tuning pipeline

Fine-tuning follows the standardized ProPicker training protocol used throughout the study. Across the UMU experiments, the main shared settings are:

- patch size 72;
- padding size 12;
- label diameter 21;
- learning rate 10^{-3};
- batch size 2 for the synthetic experiments.

Experiment 1 differs slightly because it follows the ribosome proof-of-concept regime, including a batch size of 8 and 75 epochs for the centered 128^3 crop. In the synthetic study, the incremental schedule of EXP3 adapts the number of epochs to the number of training tomograms in order to avoid unfairly undertraining the smallest subsets. The implemented schedule is $100, 80, 50, 35, 25, 22$, and 20 epochs for training sets of $1, 2, 4, 8, 12, 16$, and 20 tomograms respectively.

An important methodological detail is that all long runs are defined from the same preprocessing products, the same split definitions and the same evaluation rules before training starts. This is not cosmetic: it removes path-dependent or setup-dependent variation and ensures that differences between experiments can be interpreted as scientific differences rather than procedural ones.

3.4. Implementation details

3.4.1. Conceptual organization of the workflow

The experimental framework is organized around a small number of conceptual layers that remain stable from EXP1 to EXP4:

- a shared definition of datasets, splits and target particle identities;
- a common preprocessing convention for tomograms, coordinates and prompt crops;
- a standardized fine-tuning and inference protocol;
- a structured output space for prompts, predicted coordinates, checkpoints and localization maps.

This organization is small enough to remain transparent and strict enough to support controlled experimentation. The same conventions are reused from EXP1 to EXP4, which means that the only substantial differences between runs come from data split, prompt protocol or checkpoint choice, not from ad hoc procedural changes.

3.4.2. Monitoring, exports and visualization

The framework supports three complementary inspection levels: scalar metrics for comparison, full localization maps for qualitative review and prompt metadata tables for later robustness analysis. Localization maps can be represented both as tensor outputs for numerical post-processing and as MRC volumes for volumetric inspection in ParaView. This separation between scalar, volumetric and prompt-level evidence is important because no single inspection level is sufficient on its own to explain detector behaviour in cryo-ET.

3.5. Experimental protocol

3.5.1. Scientific workflow and reproducibility controls

The scientific workflow used in the thesis follows the same sequence in all four experiments:

1. formulate a narrowly defined experimental question and an operational hypothesis;
2. freeze the relevant split, preprocessing rules and metric implementation before launching long runs;
3. isolate the factor under study so that only one main axis changes at a time;
4. execute training or inference under standardized conditions that save checkpoints, configurations and outputs in deterministic result sets;
5. perform quantitative evaluation first and qualitative or failure analysis second.

This order is central to the credibility of the results. EXP1 changes model adaptation and inference threshold, not the dataset. EXP2 changes domain while keeping the new UMU split fixed. EXP3 changes training-set size and prompt aggregation while preserving the split and the downstream evaluator. EXP4 changes prompt orientation and checkpoint family while reusing the same validation tomograms. The framework therefore acts not only as storage, but as an experimental control mechanism.

3.5.2. Baselines and controlled comparisons

The experimental program is cumulative. Table 3.1 summarizes the completed experiments and the main comparison made in each one.

Exp.	Dataset	Goal	Main comparison
EXP1	EMPIAR-10988 ribosomes	Validate the full workflow on real data	Base ProPicker vs threshold-tuned base vs fine-tuned model
EXP2	UMU synthetic, thyroglobulin	Measure severe domain shift and recovery through adaptation	Base ProPicker vs fine-tuned model on a fixed 20/5 split
EXP3	UMU synthetic, thyroglobulin	Measure data efficiency and prompt aggregation effects	Incremental training sizes and single-prompt vs multi-prompt inference
EXP4	UMU synthetic, thyroglobulin	Study residual robustness after fine-tuning	Base vs fine-tuned families and prompt-level rotational variability

Table 3.1: Overview of the completed experiments.

3.5.3. Metrics

The main scalar metrics are

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

The study reports both aggregate metrics and per-tomogram metrics, but the geometric matching rule depends on the benchmark.

- In EXP1, the ribosome evaluation follows the cluster-based post-processing and matching criterion used in the ProPicker reference workflow. A prediction is counted as a true positive when the predicted box reaches an IoU of at least 0.6 with an expert annotation.
- In EXP2 to EXP4, the UMU evaluation matches predicted centers to ground-truth centers with a distance threshold equal to half the prompt width.
- In EXP4, prompt-level means, standard deviations and Mann–Whitney tests are added in order to determine whether family-level differences survive statistical aggregation.

Because the metrics are tied to the post-processing, calibration is part of the effective detector. In EXP1, for example, the cluster-size filters are not arbitrary technical details: they encode prior knowledge about ribosome size and therefore change the final precision-recall operating point.

3.5.4. Hyperparameter policy

The objective of the thesis is not to perform a broad hyperparameter search. Once a stable setup was found, the policy was to keep it fixed whenever possible and only change one axis of variation at a time. This decision is especially important for EXP3 and EXP4, where the point is to compare training-set size and prompt properties, not to co-optimize all training settings jointly.

The practical implication is straightforward. Patch size, padding, prompt size, contrast inversion and the UMU validation split are kept constant once they become part of the shared synthetic benchmark. Only the factors required by the research question are allowed to move: threshold calibration in EXP1, domain adaptation in EXP2, sample efficiency and prompt aggregation in EXP3, and rotational prompt diversity in EXP4.

Chapter 4

Implementation and Experimental Development

4.1. Adapting ProPicker to a reproducible workflow

One of the main practical tasks of the thesis was to move from isolated preliminary trials to a workflow that could support iterative experimentation under controlled conditions. This required more than repeating the same runs. Dataset identities, preprocessing assumptions and evaluation rules had to be stabilized so that later comparisons would remain scientifically interpretable. The resulting organization is modest in size, but it is sufficiently structured to support repeated training and inference cycles without changing the meaning of the benchmark itself.

The central design choice was to separate methodological roles clearly. Prompt selection, threshold calibration, model adaptation and failure analysis are treated as distinct stages with different scientific purposes. This separation matters because it prevents interpretability problems: a weak result can then be attributed to the prompt, the domain shift, the training-set size or the post-processing stage, rather than to an uncontrolled mixture of factors.

4.2. Prompt engineering and data preparation

Prompt construction became one of the most important technical themes of the project. At the beginning, a single prompt per particle class was enough to validate the concept. As the work progressed, prompt choice itself turned into an experimental variable.

The data-preparation stage therefore evolved in three directions:

1. **fixed prompts** for the first real and synthetic benchmarks, used to minimize confounding factors;

2. **multi-prompt aggregation** in Experiment 3, where several embeddings were combined to reduce variance and false positives;

3. **rotation-diverse prompt sets** in Experiment 4, where the study explicitly searched for prompts separated by at least 30° in orientation space and filtered them by a quality score.

This progression is meaningful because it mirrors the scientific questions of the project. The methodological machinery did not become more elaborate for its own sake. It became more elaborate because the bottleneck moved from «can the model detect anything useful?» to «why does the model still react differently to prompts that should, in principle, be equivalent?»

Two methodological decisions are worth stressing. First, prompt selection and model adaptation are treated as separate analytical stages so that prompt quality can be studied independently from learning dynamics. Second, prompt metadata are preserved alongside the selected prompt sets whenever the experiment requires later analysis. This is what makes the EXP4 robustness study possible without losing the link between prompt identity and detector behaviour.

4.3. Training and inference protocol

The work does not introduce a new neural architecture, but it does formalize the experimental pipeline needed to evaluate ProPicker systematically:

- tomograms and coordinates are prepared in a detector-compatible representation;
- training and validation conditions are fixed before model adaptation begins;
- the best-performing model state is selected under a consistent validation protocol;
- predicted localization maps are converted into coordinates and, when useful, into MRC volumes for qualitative inspection.

This formalization mattered in practice. Once Experiments 3 and 4 started producing several increments, several prompt variants and several candidate model states, informal execution would have become error-prone. The final workflow instead records those combinations in a way that is easier to repeat and audit scientifically.

4.4. Experiment-specific developments

4.4.1. EXP1: Real-data proof of concept

The first development step was to formalize the ribosome proof of concept as a controlled experiment on a fixed crop of EMPIAR-10988. The main technical addition was the explicit treatment of cluster-based post-processing as part of the experiment. Instead of treating the detector output as final, the study calibrates the binarization threshold and constrains admissible cluster sizes around the expected ribosome volume. This makes it possible to separate raw model quality from threshold calibration effects.

4.4.2. EXP2: Stable synthetic baseline

The second step introduced a cleaner synthetic benchmark. The methodology isolates label 7, corresponding to thyroglobulin, converts coordinates from Å to voxels, extracts 37^3 prompts and generates prompt embeddings under the same contrast convention used in the rest of the study. The resulting 20/5 split defines a stable single-class benchmark and becomes the anchor for all subsequent synthetic experiments.

4.4.3. EXP3: Incremental training and prompt aggregation

EXP3 required a more structured design. The study had to manage several training increments and evaluate them under different prompt strategies without changing the validation set. This is where the project stopped being a simple reproduction effort and became a genuine experimental platform. The methodology introduces three key elements: an increment schedule with adaptive epochs, a robust single-prompt selector based on the physical score q_i, and a multi-prompt representation based on the mean embedding of ten candidate particles. In practical terms, the single-prompt branch ranks candidates with the Fourier ratio, center-to-border ratio and DC-penalty terms described in Chapter 3 and then keeps the best physically plausible particle. The multi-prompt branch samples ten crops across the variance-ranked candidate list, computes ten embeddings and uses their arithmetic mean as the operational multi-instance representation. This allows the same fine-tuned model to be evaluated under single-prompt and multi-prompt conditioning without changing the validation protocol.

4.4.4. EXP4: Rotation sensitivity analysis

The fourth step introduced prompt sets with explicit rotational diversity and a dedicated robustness analysis. The selection procedure computes quality scores, removes particles that are too close to tomogram borders, reads orientation information from the annotations and then performs greedy prompt selection in orientation space. The same validation protocol is then reused for the base family and for the fine-tuned single-prompt and multi-prompt families. Finally, prompt-level metrics are correlated with positional, rotational and embedding descriptors, which turns prompt robustness into a measurable object rather than a purely qualitative concern.

4.5. Current state of the experimental framework

As of March 2026, the project contains completed experimental developments for EXP1 to EXP4 and a written specification for EXP5 multiclass fine-tuning. The framework supports:

- prompt extraction and prompt embedding generation;
- fine-tuning on real and synthetic datasets;
- validation-time inference with consistent model-state selection;
- export of predicted coordinates and volumetric localization maps;
- statistical and qualitative failure analysis.

What is still missing is not infrastructure, but the next round of experiments. In particular, the multiclass benchmark described in the documentation has not yet been executed and therefore remains future work.

Chapter 5

Experiments, Results, and Discussion

5.1. Experimental setup

The four completed experiments form a coherent sequence rather than a collection of isolated tests. EXP1 establishes real-data viability. EXP2 measures the impact of strong domain shift. EXP3 quantifies how much annotated data is needed and whether prompt aggregation is worthwhile. EXP4 investigates the residual source of instability after fine-tuning has already solved the large domain-shift problem.

Across the synthetic experiments, three protocol decisions are especially important:

1. the validation set remains fixed;
2. post-processing is reused consistently;
3. changes are introduced one axis at a time, such as training-set size or prompt strategy.

5.2. Experiment 1: EMPIAR-10988 ribosomes

EXP1 was designed as an initial validation on real cryo-ET data. The goal was to test whether the pretrained detector could already identify a meaningful signal on ribosomes and whether a lightweight fine-tuning stage could convert that signal into a balanced detector.

5.2.1. Question and hypothesis

The scientific question of EXP1 is whether ProPicker contains enough transferable prior information to detect ribosomes on real cryo-ET data before any task-specific adaptation. The working hypothesis is deliberately moderate: the pretrained detector should show useful signal, but that signal will likely be poorly calibrated and biased toward precision rather than recall.

5.2.2. Implementation and controls

The experiment uses the `cyto_ribosome` case of EMPIAR-10988, with `TS_029` for training and `TS_030` for validation under `crop_delta=64`. The methodology is carried out in two stages. First, the base model is run in prompt-only mode and its localization map is transformed into coordinates through cluster post-processing. This post-processing is treated explicitly rather than as an opaque default. It calibrates:

- the binarization threshold, which controls the precision-recall balance through the size of the activated clusters;
- the minimum cluster size, initialized near 0.1 times the expected ribosome ball volume;
- the maximum cluster size, initialized near 1.2 times the same volume.

The evaluator then applies the same criterion used in the reference workflow, counting a detection as correct when it reaches IoU ≥ 0.6. Second, the model is fine-tuned on the cropped ribosome setting with the same prompt construction and the same evaluation rule. This preserves the scientific comparison: only the model adaptation changes, not the data split or the evaluator.

Setting	Precision	Recall	F_1	TP	FP	FN
Base ProPicker	0.7049	0.1966	0.3074	547	229	2236
Base with optimized threshold	0.5125	0.2648	0.3492	737	701	2046
Fine-tuned model	0.5677	0.5752	0.5714	1044	795	771

Table 5.1: EXP1 results on EMPIAR-10988.

Table 5.1 shows the key finding of the real-data experiment. The pretrained model is not useless; it is conservative. It finds ribosomes with decent precision, but its recall is too low for operational use. Threshold tuning improves F_1 slightly, which confirms that part of the issue is calibration. Still, the decisive improvement comes from fine-tuning: the F_1 score rises from the 0.31 to 0.35 range into 0.5714, with recall more than doubling relative to the base model.

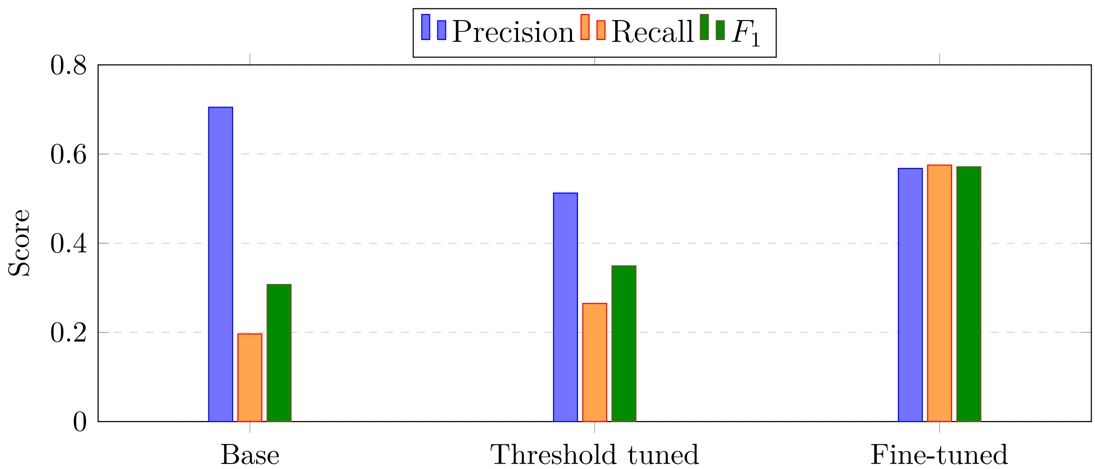

Figure 5.1: EXP1 metric balance. Threshold tuning changes calibration, but fine-tuning is the only step that raises recall and F_1 together.

5.2.3. Interpretation

The result justifies the rest of the thesis. Without it, one could argue that the problem lies entirely in a broken pipeline. Instead, EXP1 demonstrates three things. First, the promptable formulation carries useful information even on real data. Second, threshold calibration matters enough to deserve explicit treatment. Third, calibration alone does not close the gap: the real change of regime comes from fine-tuning, which lifts recall from 0.1966 to 0.5752 and reduces the false-negative count from 2236 to 771. Figure 5.1 makes that rebalancing visible at a glance. In scientific terms, the experiment validates the pipeline and isolates the first strong conclusion of the thesis: prompt-only transfer is informative, but insufficient for balanced particle picking on real ribosome data.

5.2.4. Direct summary

For a non-specialist reader, EXP1 says something simple: ProPicker already *sees* ribosomes in real data, but in its pretrained state it misses too many of them to be trusted as a final detector. Fine-tuning is the step that converts that weak but real signal into a usable operating point. The main limit observed here is therefore not total failure, but lack of balance between precision and recall before adaptation.

5.3. Experiment 2: UMU synthetic domain shift and recovery

EXP2 asks a sharper question: what happens when the detector is moved to a synthetic domain that differs strongly from the conditions under which the pretrained model is useful?

5.3.1. Question and hypothesis

The question is whether the signal observed in EXP1 survives a much stronger domain shift. The hypothesis is that the pretrained model will collapse on UMU synthetic thyroglobulin, but that fine-tuning on a controlled split should recover high performance if the rest of the pipeline is coherent.

5.3.2. Implementation and controls

The experiment freezes the now-standard 20/5 split of the 25 UMU tomograms and keeps that validation set unchanged in all later synthetic experiments. The preprocessing stage performs the following transformations in a reproducible order:

1. reconcile annotation metadata with the local identity of each tomogram;
2. retain only label 7, corresponding to `thyroglobulin`;
3. convert particle coordinates from Å to voxels;
4. extract $37 \times 37 \times 37$ prompts;
5. invert contrast to match the ProPicker/TomoTwin convention;

6. generate the normalized inputs required by the detector.

The evaluation stage uses the same validation tomograms in all comparisons and matches predicted centers to ground truth with a threshold of half the prompt size. This is a simple but crucial control: the experiment compares models under a fixed geometric rule rather than under manually adjusted tolerances.

Validation tomogram	Precision	Recall	F_1	TP	FP	FN
`tomo_rec_5_snr1.66`	0.894	1.000	0.944	127	15	0
`tomo_rec_6_snr1.17`	0.895	0.992	0.941	119	14	1
`tomo_rec_7_snr1.13`	0.868	0.985	0.923	131	20	2
`tomo_rec_8_snr0.57`	0.565	0.939	0.706	108	83	7
`tomo_rec_9_snr1.28`	0.888	1.000	0.941	127	16	0
Mean fine-tuned performance	0.822	0.983	0.891	–	–	–
Mean base performance	0.009	0.008	0.008	–	–	–

Table 5.2: EXP2 results on the fixed UMU synthetic validation set.

The contrast in Table 5.2 is the strongest result of the thesis. The pretrained model essentially fails on this domain, while the fine-tuned model reaches mean $F_1 = 0.891$. This is not a subtle effect. It establishes that under substantial domain shift, fine-tuning is not an optional refinement but the condition that makes the method usable.

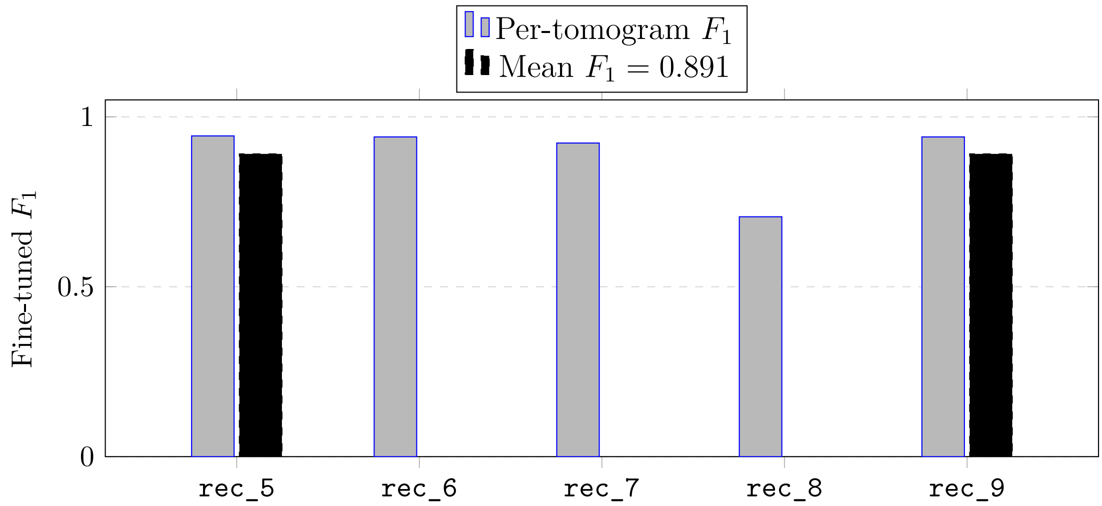

Figure 5.2: EXP2 fine-tuned performance by validation tomogram. The noisiest validation volume, `tomo_rec_8_snr0.57`, is the only clear outlier.

5.3.3. Interpretation

The table also exposes the main failure mode that remains after adaptation. Performance is strong and consistent in four tomograms, but `tomo_rec_8_snr0.57` is visibly harder, as Figure 5.2 makes clear. The drop is driven mainly by precision, not by recall. This matters scientifically because it changes the interpretation of the residual error: after domain adaptation, the dominant issue is no longer a failure to activate detections at all, but a loss of selectivity under difficult noise conditions. That observation becomes a recurring theme in the next experiments.

5.3.4. Direct summary

In plain language, EXP2 marks the first hard limit of ProPicker. The pretrained detector does not transfer automatically to UMU thyroglobulin and is essentially unusable without adaptation. Once the model is fine-tuned, however, the same pipeline becomes strong and stable on four of the five validation tomograms. From this point onward, the problem is no longer «the model detects nothing», but «the model becomes too permissive in the hardest tomograms».

5.4. Experiment 3: Data efficiency and multi-prompt inference

EXP3 keeps the same validation benchmark and varies the number of training tomograms. It also compares single-prompt and multi-prompt inference.

5.4.1. Question and hypothesis

The central question of EXP3 is practical: how much annotated data is needed to recover most of the attainable performance, and does prompt aggregation provide a measurable advantage over a single prompt? The hypothesis is that the learning curve will rise sharply at small sample sizes and then saturate, while multi-prompt inference will reduce false positives by averaging away prompt-specific noise.

5.4.2. Implementation and controls

The experiment reuses exactly the same validation set as EXP2 and trains seven checkpoints with $1, 2, 4, 8, 12, 16$, and 20 training tomograms. The adaptive-epoch policy ensures that small subsets receive longer optimization and are not penalized by construction.

The experiment also introduces the first fully explicit prompt-selection strategy of the thesis. The single-prompt branch uses a robust quality score based on three interpretable terms:

- a Fourier mid-band to high-band energy ratio, which favors structured signal over high-frequency noise;
- a center-to-border ratio, which favors crops whose main mass is centered;
- a DC penalty, which discourages crops dominated by offset or gradient artifacts.

The multi-prompt branch selects ten diverse crops by variance ranking, encodes them with TomoTwin and represents the class through both the first embedding and the mean embedding. The resulting prompt statistics show that the cosine similarity between the single and multi representations is 0.9629, with mean embedding-dimension variance 0.010789. This is important because it shows that the two prompt representations correspond to closely related instances of the same particle class; the expected gain is therefore robustness, not a change of class semantics.

Training tomograms	F_1 single prompt	F_1 multi-prompt $n = 10$
1	0.691	0.647
2	0.728	0.717
4	0.800	0.875
8	0.860	0.906
12	0.862	0.919
16	0.888	0.911
20	0.849	0.894

Table 5.3: EXP3 mean F_1 scores as a function of training-set size and prompt strategy.

The shape of Table 5.3 is more informative than any single number. Both strategies improve rapidly when a few tomograms are added. From 4 tomograms onward, the multi-prompt strategy becomes clearly superior. The best observed configuration is reached at 12 tomograms with multi-prompt inference, obtaining $F_1 = 0.919$. After that point, the curve saturates and even fluctuates slightly downward.

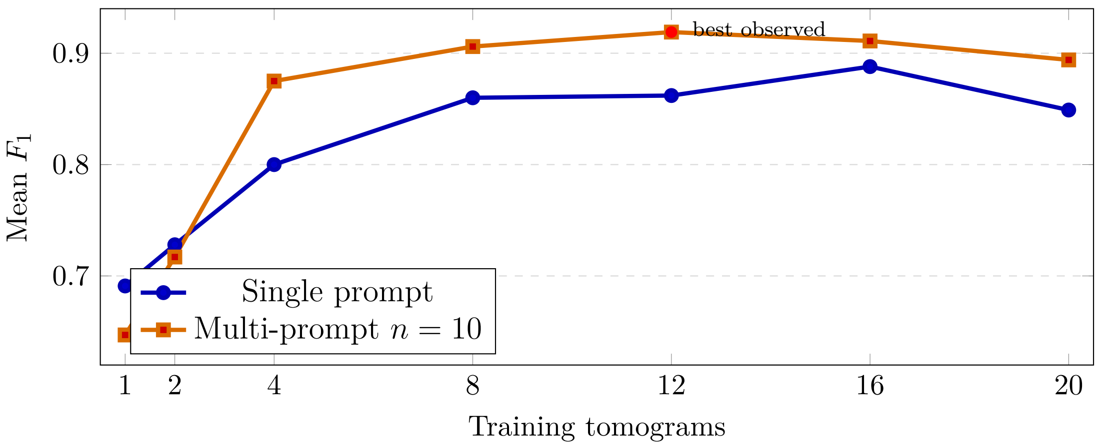

Figure 5.3: EXP3 learning curves. Most of the gain appears early, and the best observed configuration is the multi-prompt model trained with 12 tomograms.

5.4.3. Interpretation

This result has direct practical value. If annotation time is limited, 4 tomograms already define a strong operating point. If the priority is peak performance under the current protocol, 12 tomograms with multi-prompt inference are the best choice. Figure 5.3 makes the rapid rise and subsequent plateau explicit. The reason why multi-prompt wins is not that it recovers more particles at all costs. It wins because it suppresses false positives much more effectively. At the 12-tomogram setting, the single-prompt model yields 190 false positives, while the multi-prompt model reduces them to 70, with only a moderate loss in true positives.

The implication is important for cryo-ET workflows. A detector with slightly lower recall but far fewer false positives may be operationally preferable because it reduces manual curation and prevents contamination of downstream subtomogram processing. Methodologically, EXP3 also shows that prompt selection cannot remain an informal choice once the detector is already strong. The implementation of prompt ranking and aggregation becomes part of the model-selection problem itself.

5.4.4. Direct summary

The message of EXP3 is that the optimized pipeline does not need a large labeled dataset to become useful. With only 4 training tomograms, ProPicker already enters a high-performance regime, and with 12 tomograms plus multi-prompt inference it reaches the best observed result of the whole study. The benefit of the multi-prompt strategy is practical and easy to explain: it makes the detector less impulsive, so it raises precision by rejecting many false alarms without destroying recall.

5.5. Experiment 4: Prompt-robustness analysis

EXP4 studies the residual instability that remains after fine-tuning, but the analysis is structured around a more precise question than in the earlier exploratory version. The issue is not merely whether a prompt has a «bad rotation». The relevant question is whether failures are better explained by raw tomogram quality, by acquisition anisotropy from the missing wedge, or by symmetry-aware prompt ambiguity in the specific case of thyroglobulin. In other words, EXP4 is the point where the thesis moves from a generic prompt-robustness study to a physically grounded interpretation of prompt failure.

5.5.1. Question and hypothesis

The question is whether a strong fine-tuned checkpoint is effectively invariant to prompt identity once the domain-shift problem has been solved, and if not, what kind of physical explanation best fits the remaining failures. The working hypothesis is threefold. First, family-level averages after fine-tuning can still look strong even when some prompts are unusable. Second, the dominant failure mode will be recall collapse rather than a uniform increase in false positives. Third, simple source-tomogram SNR or a single angle-to-Z explanation will be too weak to account for the worst prompts; the more realistic explanation will involve the interaction between prompt representativeness, missing-wedge anisotropy and the C2 symmetry of thyroglobulin.

5.5.2. Implementation and controls

The current EXP4 analysis is based on the large prompt-population notebook and deliberately uses only the training pool to avoid any validation leakage into prompt construction. A population of 200 prompts is generated, and the first 125 prompts are used as the statistical study subset. The focus checkpoint is `multi_inc16`, chosen because EXP3 showed it to be one of the strongest and most stable operating points. Each prompt is evaluated on the fixed 5-tomogram validation set, and the analysis is carried out on prompt-level mean metrics aggregated over those validation tomograms.

Prompt selection is explicit and staged:

1. remove particles that are closer than 50 voxels to the tomogram borders;

2. compute the same physical-quality score introduced in EXP3;

3. keep only the top 35% quality candidates, implemented as a quality-percentile threshold of 65;
4. build a rotation-diverse prompt population by greedy quaternion-distance selection;
5. restrict the statistical analysis to prompt indices 0 to 124.

The descriptors extracted for each prompt are theory-guided rather than purely ad hoc. Quaternion metadata are converted into orientation features relative to the global Z axis, which serves as a proxy for beam-axis and tilt-axis sensitivity. The prompt subtomogram itself is characterized through source SNR, centering statistics, Fourier-band ratios, a missing-wedge anisotropy proxy, mass-center shift, inertia anisotropy and embedding descriptors. Prompt position inside the source tomogram is also retained, including normalized Z coordinate, distance to the tomogram center and distance to the borders. This produces an analysis aligned with the cryo-ET discussion of Section 2.1.1: not pure rotation in SO(3), but rotation under anisotropic acquisition and object symmetry.

Quantity	Value
Study subset	First 125 prompts from the 200-prompt population, all extracted from training tomograms only
Focus checkpoint	`multi_inc16`, used as the operational checkpoint for detailed prompt analysis
Global prompt-level performance	$F_1 = 0.780 \pm 0.283$, Precision = 0.824, Recall = 0.783
Best observed case	Prompt 80 on `tomo_rec_6_snr1.17`, $F_1 = 0.983$
Hardest validation tomogram	`tomo_rec_8_snr0.57`, mean $F_1 = 0.659$ across the 125 prompts
Dominant failure pattern	Eight prompts reach mean recall = 0.0; prompt 124 reaches Precision = 1.000 but Recall = 0.115, showing severe under-activation rather than noisy overprediction

Table 5.4: Summary of the restructured EXP4 analysis based on the 125-prompt study subset.

Factor	Signal on mean recall	Interpretation
Source prompt SNR	$r = +0.051$	Prompt-source tomogram quality alone does not explain the worst failures
C2-axis angle to global Z	$r = +0.061$	The descriptive direction is compatible with the missing-wedge story, but it is too weak to act as a stand-alone rule
Missing-wedge anisotropy proxy	$r = +0.082$	Acquisition anisotropy matters conceptually, but this scalar proxy alone is not dominant
Distance to tomogram center	$r = +0.195$, $p = 0.0295$	Positional effects inside the source tomogram are detectable
Edge-distance measure	$r = -0.253$, $p = 0.0044$	Spatial context and prompt placement are more informative than source SNR
Embedding dispersion (`emb_std`)	$r = +0.146$, $p = 0.1051$	Embedding-space spread is suggestive, but not strong enough on its own

Table 5.5: Selected signals from the 125-prompt EXP4 analysis. Correlations are reported against mean recall, which is the main failure indicator in this experiment.

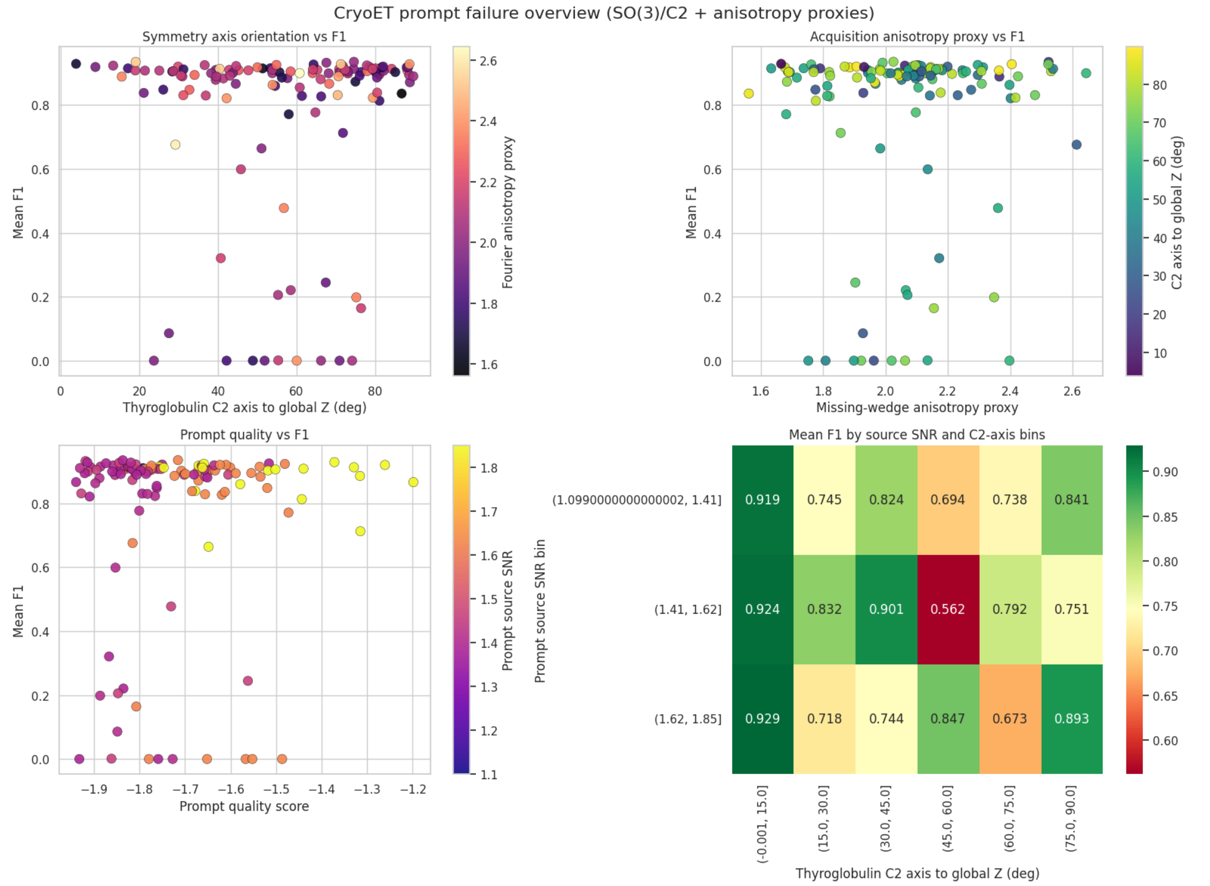

Figure 5.4: Overview of the final EXP4 prompt-level analysis. The scatter plots show broad dispersion and only weak monotonic structure for orientation, anisotropy and quality features, while the bottom-right heatmap reveals patchy interactions between source-SNR bins and symmetry-axis bins rather than a single stable trend.

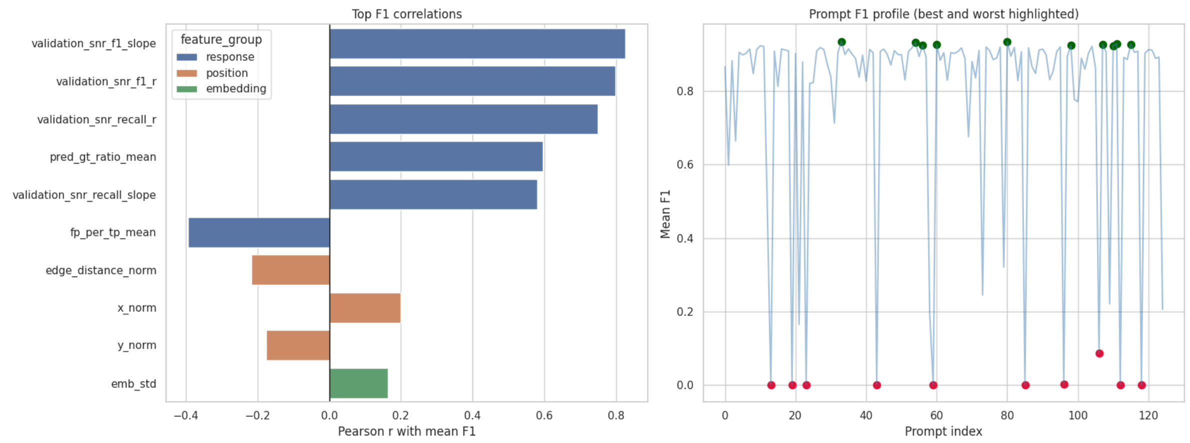

Figure 5.5: Left: the strongest raw correlations with prompt-level mean F_1 are dominated by response-derived descriptors such as validation-SNR slope and predicted-to-ground-truth count ratio. Right: prompt performance over the 125-prompt subset shows a broad high-performance plateau interrupted by a small number of catastrophic failures.

5.5.3. Interpretation

The first conclusion is that prompt failure is real and large even after fine-tuning. Table 5.4 and Figure 5.5 show a broad high-performance plateau together with a small set of catastrophic prompt failures. The failure is dominated by recall collapse. The worst prompts do not usually produce a flood of false positives; instead, they fail to activate detections at all. Prompt 124 is a particularly informative example because it reaches perfect mean precision but very low mean recall. That is the signature of an overly restrictive or non-representative template, not of a prompt that is merely noisy.

The second conclusion is that «tomogram quality» must be split into two distinct effects. The quality of the *target validation tomogram* clearly matters: `tomo_rec_8_snr0.57` is substantially harder than the other four validation tomograms. However, the quality of the *source tomogram from which the prompt was extracted* is not sufficient to explain the worst prompts. Source SNR is almost uncorrelated with mean recall, and several catastrophic prompts originate from moderate source SNR values rather than from uniquely poor tomograms. Therefore, the dominant issue is not simply that the bad prompts come from bad source volumes.

The third conclusion is that the simple geometric hypothesis «the problematic prompts are those aligned badly with the global Z axis» is not supported as a dominant explanation. The overview plots in Figure 5.4 show at most weak monotonic trends for both orientation and missing-wedge proxies. Descriptive correlations exist in the expected direction for some variables, including the C2-axis angle to global Z, but they are weak and statistically unconvincing in the 125-prompt study. This does not invalidate the acquisition theory presented in Section 2.1.1; rather, it shows that the acquisition effect interacts with other factors and is not recoverable through one naive angle feature alone.

This is where thyroglobulin symmetry becomes important. Because the target is a C2 homodimer, the physically meaningful pose space is $SO(3)/C_2$, not pure $SO(3)$ [18]. Some prompt views are therefore intrinsically less informative because they lie near symmetry-equivalent configurations. When this intrinsic ambiguity is combined with missing-wedge anisotropy and with whatever local context is contained in the crop, the resulting prompt can become a poor class representative even if its raw SNR is acceptable. The worst prompts are better understood as symmetry-aware, acquisition-amplified ambiguous templates than as merely low-quality tomographic patches.

The positional features in Table 5.5 reinforce that interpretation. Distance-to-center and edge-distance effects are more visible than source SNR. This suggests that prompt representativeness depends not only on nominal particle orientation, but also on the spatial context from which the crop is extracted: centering, local neighborhood and residual reconstruction bias inside the source tomogram. Under this reading, EXP4 is not a verdict that «rotation does not matter»; it is a more precise verdict that rotation matters through a compound mechanism involving symmetry, anisotropic acquisition and prompt content.

The fourth conclusion is that the stronger-looking response features should be interpreted as diagnostics, not as causes. Figure 5.5 shows that the largest raw correlations with mean F_1 come from prompt-response summaries such as the validation-SNR slope and the predicted-to-ground-truth count ratio. These variables are useful because they reveal which prompts generalize stably across the five validation tomograms, but they are downstream reflections of prompt quality rather than upstream explanations.

The fifth conclusion is that the rotation-plus-acquisition story does not survive adjustment as a dominant model. The adjusted regressions add only $\Delta R^2 = 0.0109$ for mean F_1 and $\Delta R^2 = 0.0252$ for mean recall once prompt quality, position terms and source-tomogram effects are already in the model. The tomogram-stratified permutation test confirms that these gains are fully compatible with the null distribution. In parallel, the PCA and clustering analysis does not reveal a clean bad-prompt cluster: the best two-cluster solution has a silhouette score of only 0.225 and both clusters have almost identical mean F_1 values (0.779 and 0.781). The simplest interpretation is that prompt failure is sparse and heterogeneous, not a neatly separated prompt subtype.

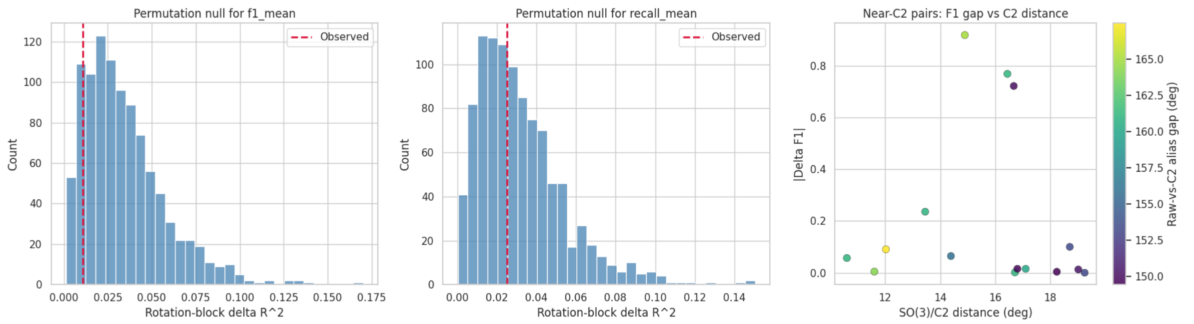

Figure 5.6: Causal checks for the final EXP4 analysis. The observed contribution of the rotation-plus-acquisition block lies well inside the tomogram-stratified permutation null for both mean F_1 and mean recall. The right panel shows that even near-$\mathrm{SO}(3)/\mathrm{C}_2$ prompt pairs can still exhibit substantial F_1 differences, which is consistent with prompt ambiguity being shaped by more than raw pose distance alone.

The C2-consistency diagnostics make that last point concrete. Among the 15 prompt pairs that are close in $\mathrm{SO}(3)/\mathrm{C}_2$ ($\leq 20°$), the mean absolute F_1 gap is still 0.2005, with a few very large outliers. Therefore, even prompts that are nearly equivalent after symmetry collapse can behave very differently once acquisition footprint and crop content are included. A bad prompt is thus best defined operationally: it is a prompt that causes recall collapse on held-out tomograms, not a prompt that can be identified reliably from source SNR or from one orientation angle alone.

5.5.4. Direct summary

The important lesson of EXP4 is that good average performance can hide catastrophic prompt failures. After fine-tuning, ProPicker is no longer limited mainly by lack of training data. It is limited by the fact that some prompts are poor representatives of the particle class. The present evidence does not support the claim that this is explained by source SNR alone, nor by a single angle-to-Z rule, nor by the rotation-plus-acquisition block as a whole once confounders are controlled. The most defensible interpretation is narrower and more useful: a bad prompt is one that causes held-out recall collapse because its representation is not robust to the joint effects of C2 symmetry, anisotropic acquisition and local crop context. The natural next steps are therefore symmetry-aware evaluation in $\mathrm{SO}(3)/\mathrm{C}_2$, missing-wedge-matched data augmentation and explicit prompt-quality screening based on representativeness rather than on SNR alone.

5.6. Cross-experiment discussion

The four experiments support a cumulative interpretation.

First, prompt-only transfer is real but incomplete. EXP1 shows that the pretrained detector already contains useful information on real data. Second, severe domain shift can still break the method almost completely, as EXP2 demonstrates on UMU thyroglobulin. Third, once fine-tuning is introduced, most of the performance can be recovered with relatively little annotated data, and multi-prompt inference provides a practical precision gain, as shown in EXP3. Fourth, after the major domain-adaptation problem has been solved, the dominant unresolved limitation becomes prompt sensitivity, as established by EXP4.

This sequence defines the current state of the investigation very clearly. The project is no longer asking whether the general strategy works at all. That question has been answered positively. The open question is now how to make the strategy stable enough to justify the word «universal» in a stronger sense.

5.7. Extended summary for non-expert readers

Table 5.6 condenses the completed experiments into the simplest possible message: where ProPicker already works, where it still breaks and what the optimized pipeline should do about it.

Experiment	Main result	Practical meaning
EXP1	Base ProPicker already detects ribosomes, but with $F_1 = 0.3074$; fine-tuning raises the result to $F_1 = 0.5714$	The base model is informative on real data, but balanced use still requires adaptation
EXP2	On UMU thyroglobulin the pretrained model collapses ($F_1 \approx 0.008$); fine-tuning reaches mean $F_1 = 0.891$	Severe domain shift is the first hard limit of ProPicker and must be corrected with fine-tuning
EXP3	The best observed configuration is 12 tomograms with multi-prompt inference ($F_1 = 0.919$); 4 tomograms already define a strong regime	The current pipeline is data-efficient and multi-prompt conditioning is the best default operating mode
EXP4	Good averages hide recall-collapse prompts; source SNR and simple rotation rules are too weak on their own, and the adjusted rotation/acquisition block is not dominant	The main remaining limit is prompt representativeness under symmetry and anisotropic acquisition, not lack of training data

Table 5.6: Plain-language synthesis of the completed experiments.

Viewed end to end, the results answer the central project objective directly. ProPicker is already strong enough to justify serious experimentation, but not strong enough to be called universally reliable without qualification. Its first practical limit is domain shift: the pretrained model does not carry over cleanly to a new benchmark. Its second limit is finer but just as important: even after successful fine-tuning, the detector still depends too much on which prompt is used to represent the target particle.

The optimized pipeline proposed in this memory follows from that diagnosis. First, enforce data consistency and voxel-space annotations. Second, assume that fine-tuning will be required whenever the target domain changes substantially. Third, use a quality-controlled multi-prompt representation as the default inference mode once a particle-specific checkpoint exists. Fourth, stop judging the system only by a single mean score and inspect prompt-level failures explicitly. Under the current evidence, this is the shortest path from a promising promptable detector to a dependable working workflow.

5.8. Pipeline optimization and recommendations

The main practical objective of the project is not only to report benchmark scores, but to optimize a usable and reproducible ProPicker pipeline for cryo-ET particle picking. Once EXP1 to EXP4 are interpreted together, they yield a concrete set of engineering recommendations. The central message is that the pipeline should not be optimized by chasing isolated hyperparameters. It should be optimized in stages, starting from data consistency, then prompt quality, then fine-tuning strategy, and only afterwards inference calibration and robustness analysis.

5.8.1. Recommended baseline workflow

The recommended workflow that emerges from the investigation is the following:

1. freeze the dataset split and coordinate convention before running any model;
2. preprocess tomograms into a toolchain-compatible layout, including contrast inversion when required and coordinate conversion into voxels;
3. build prompts from physically plausible particles rather than from arbitrary crops;
4. fine-tune the model as soon as a substantial domain shift is present;
5. evaluate under a fixed metric implementation and store both scalar metrics and volumetric outputs;
6. inspect errors by tomogram and by prompt before changing the next factor.

This workflow is stricter than a typical exploratory analysis cycle, but the experiments show why that strictness is necessary. If the split moves, EXP2 to EXP4 cease to be comparable. If prompt construction is informal, EXP3 and EXP4 become partly uninterpretable. If post-processing is hidden, as would have happened in EXP1, calibration effects become impossible to separate from model adaptation.

5.8.2. Optimization priorities derived from the experiments

The first optimization priority is **data and annotation consistency**. The UMU experiments show that coordinate conversion and split stability are not administrative details. They are the basis on which every later comparison rests. The pipeline should therefore enforce consistent tomogram identities, voxel-space coordinates, a stable contrast convention and a fixed validation subset once a benchmark is established.

The second priority is **prompt quality control**. EXP3 and EXP4 make clear that prompt choice cannot be left to convenience. A good operational prompt should satisfy three conditions: it should contain structured mid-frequency content, be centered with respect to the particle mass, and avoid strong DC or border artifacts. When the goal is deployment rather than diagnosis, the best default choice is the multi-prompt strategy, because it stabilizes the representation and reduces false positives. When the goal is robustness analysis, the prompt set should instead be explicitly diversified in orientation space so that failure modes become observable.

The third priority is **fine-tuning policy**. EXP2 demonstrates that fine-tuning is mandatory under strong domain shift, and EXP3 shows that the return on added supervision is highly non-linear. The recommended strategy is therefore staged: start with a small but carefully prepared subset, evaluate early, and only expand the training pool if the new data improve the validation set under the same prompt protocol. In the present benchmark, 4 training tomograms already define an efficient operating point and 12 tomograms with multi-prompt inference define the best observed compromise between annotation cost and peak performance.

The fourth priority is **inference calibration and post-processing**. EXP1 shows that thresholding and cluster filtering are not secondary. They are part of the effective detector. The pipeline should expose these parameters explicitly, document their rationale and tune them under a fixed evaluator. For real data, cluster-size priors tied to the expected particle volume are especially useful. For synthetic center-based evaluation, the distance threshold should remain attached to the prompt size so that later comparisons are geometrically consistent.

The fifth priority is **failure analysis as a first-class step**. EXP4 shows that a high mean F_1 can hide fragile prompts. For that reason, the optimized pipeline should not stop at aggregate metrics. It should also store prompt metadata, report per-tomogram performance, and inspect localization maps in ParaView when numerical results alone do not explain a failure. The project learned that prompt robustness is not a cosmetic add-on. It is one of the last barriers between a promising detector and a dependable one.

Pipeline stage	Recommendation	Experimental basis
Data preparation	Enforce local-path rewriting, voxel-space coordinates, contrast convention and fixed validation splits	EXP2 establishes the stable UMU benchmark used again in EXP3 and EXP4
Prompt construction	Replace ad hoc prompt choice with quality scoring and prefer aggregated prompts for operational inference	EXP3 shows that multi-prompt conditioning reduces false positives and improves the best F_1
Fine-tuning	Treat adaptation as mandatory under domain shift and expand training data incrementally rather than all at once	EXP2 shows collapse of the base model and EXP3 shows early performance saturation
Inference calibration	Expose thresholds and cluster-size rules explicitly instead of treating them as implicit procedural choices	EXP1 separates threshold effects from true adaptation effects
Robustness analysis	Evaluate prompts individually and relate failures to symmetry, acquisition geometry and spatial context, not to SNR alone	EXP4 shows that prompt variability survives after fine-tuning and is not explained by source SNR or a single angle feature alone
Visualization and auditing	Export localization maps and keep outputs organized by experiment, prompt and checkpoint family	All experiments rely on consistent experimental organization to make comparisons and qualitative inspection tractable

Table 5.7: Recommended pipeline optimizations derived from the four completed experiments.

5.8.3. Operational recommendations

From an operational perspective, the optimized pipeline should follow a simple decision rule. If the task is a new particle in a new domain, do not rely on prompt-only inference as the final solution. Use it only as an initial signal check. If labeled data are available, move quickly to fine-tuning. If annotation budget is scarce, begin with a compact subset and measure whether the model has already entered the high-performance regime. If false positives dominate, prefer multi-prompt inference before collecting substantially more data. If performance still depends strongly on the prompt, move to quality-controlled or rotation-diverse prompt sets and perform prompt-level diagnostics.

This recommendation hierarchy reflects the actual empirical order of gains observed in the thesis. The largest benefit comes from domain adaptation. The second largest benefit comes from disciplined prompt handling. The remaining improvements come from calibration and robustness control. In other words, the pipeline should be optimized from coarse corrections to fine corrections, not the other way around.

5.9. Limitations

The thesis should be read with several limitations in mind:

- only one real-data proof of concept is included, and it uses a controlled crop rather than a full-dataset benchmark;
- the completed synthetic study is single-class, even though the next multiclass benchmark is already specified;
- some evaluation thresholds are inherited from the original reference protocol and are not independently re-derived in every experiment report;
- the repository documents the experimental GPU clearly, but not every hardware parameter of the training workstation was preserved;
- prompt selection is itself a source of variance, which means that reported scores should be interpreted together with the prompt protocol used to obtain them.

Chapter 6

Conclusions and Future Work

6.1. Conclusions

The thesis supports four final conclusions.

First, ProPicker is already a useful research platform for flexible cryo-ET particle picking. EXP1 shows that prompt-only transfer is informative even on real data, and EXP2 shows that the same platform can be specialized effectively through fine-tuning once a stable benchmark is defined.

Second, universality is currently conditional. The pretrained detector alone does not survive substantial domain shift. On UMU thyroglobulin, the base model collapses to $F_1 \approx 0.008$, whereas fine-tuning restores the detector to a strong operating regime. In practice, this means that prompt-only inference is a useful signal check, not a final deployment strategy, whenever the domain changes substantially.

Third, the current pipeline is data-efficient. A strong fraction of the attainable synthetic performance is reached with only a small subset of the training tomograms. The best observed operating point is 12 training tomograms with multi-prompt inference, while 4 tomograms already provide a compelling trade-off between annotation cost and quality. This is one of the strongest positive results of the thesis, because it shows that adaptation is feasible without a large annotation campaign.

Fourth, the most important unresolved problem is no longer gross adaptation but prompt robustness. The final EXP4 analysis shows that bad prompts are best defined operationally as prompts that trigger recall collapse on held-out tomograms. Source SNR alone does not explain them, simple angle-to-Z rules do not explain them, and the adjusted rotation-plus-acquisition block does not dominate after confounder control. The remaining barrier is prompt representativeness under the joint effects of symmetry, anisotropic acquisition and spatial context. For this reason, the current detector is best described as *adaptable and promising*, but not yet universally reliable in the strong sense implied by the thesis title.

Taken together, these conclusions answer the central research objective of the thesis. Universal cryo-ET detectors are not a solved problem today, but promptable detectors are already strong enough to justify serious experimental use when they are embedded in a disciplined workflow. The thesis therefore closes with a qualified but positive statement: ProPicker is not yet a universal detector, but it is already a credible foundation on which a more universal detector can be built.

6.2. Future directions

The next steps follow directly from the evidence produced in this thesis:

1. convert EXP4 into a practical prompt-screening stage by testing whether prompt-quality models can identify recall-collapse prompts before inference;
2. execute the planned EXP5 multiclass benchmark on UMU synthetic data using the same fixed train/validation split and the three selected classes: `thyroglobulin`, `pdb_4v94` and `pdb_4cr2`;
3. extend the real-data evaluation beyond the current ribosome proof of concept to determine whether the same trends hold across more particles and more acquisition settings;
4. investigate symmetry-aware data augmentation, SO(3)/C_2-aware evaluation and prompt-representativeness objectives that are better aligned with the physics of cryo-ET acquisition;
5. study calibration and post-processing strategies that preserve recall while reducing false positives in difficult tomograms.

If those directions succeed, the project will move from a strong single-class promptable detector toward a more general and deployable multiclass cryo-ET localization system.

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Appendix A

Hardware Specifications and Software Stack

A.1. Documented experimental hardware

Table A.1 summarizes the hardware information of the workstation used in inference and fine-tuning tasks in the AIKE laboratory.

Component	Documented value
Main training GPU	NVIDIA GeForce RTX 4090
Execution style	Linux workstation with CUDA-enabled environment
External data storage	SSD-mounted EMPIAR-10988 location
CPU	Not preserved in the repository metadata of the laboratory workstation
RAM	Not preserved in the repository metadata of the laboratory workstation
Additional metadata	CUDA-capable Linux environment documented; exact driver and CUDA toolkit versions were not archived

Table A.1: Experimental hardware documented by the repository.

A.2. Local authoring and validation environment

Part of the writing and repository inspection for this memory was carried out on the local system shown in Table A.2. This machine should not be confused with the documented experimental training workstation.

Component	Value
Computer	MacBook Pro (`Mac14,10`)
Processor	Apple M2 Pro, 12 CPU cores (8 performance + 4 efficiency)
Memory	16 GB unified memory
Integrated GPU	Apple M2 Pro, 19 GPU cores
Operating environment	macOS workstation used for repository editing and manuscript preparation

Table A.2: Local authoring environment used for the present manuscript work.

A.3. Software, languages and technologies

Table A.3 summarizes the software stack and the technologies used in the project.

Category	Tool or technology	Role in the project
Programming language	Python	Main implementation language for preprocessing, fine-tuning, inference and analysis scripts.
Programming language	LaTeX	Preparation of the thesis manuscript and final report.
Data and config formats	JSON, CSV, MRC	Prompt embeddings, coordinate tables and tomogram/localization-map storage.
Interactive environment	Jupyter Notebook	Exploratory preprocessing, metric analysis, prompt inspection and plotting.
Core detector	ProPicker	Promptable 3D segmentation model used as the central detector platform.
Training support	`DeepETPicker_ProPicker` scripts	Generation of preprocessing configs, training configs and command-line training/inference execution.
Prompt encoder	TomoTwin model	Extraction of prompt embeddings from $37 \times 37 \times 37$ subtomograms.
Deep learning library	PyTorch	Checkpoint loading, tensor manipulation and inference-time handling of localization maps.
Scientific Python	NumPy and Pandas	Numerical operations, table manipulation and result aggregation.
Visualization	Matplotlib and Seaborn	Metric plots, prompt-distribution analysis and comparative figures in notebooks.
Scientific analysis	SciPy	Distance computation, rotational analysis, statistics, FFT-based inspection and image-processing utilities.
Dimensionality reduction	scikit-learn	PCA and t-SNE analyses used in prompt-embedding exploration.
Volume I/O	`mrcfile` and toolchain MRC utilities	Reading and writing tomograms, labels and localization maps.
Logging analysis	TensorBoard event accumulator	Extraction of training curves and run statistics from event files.

Category	Tool or technology	Role in the project
Visualization of volumes	ParaView	Inspection of tomograms and exported localization maps in volumetric form.
Environment management	Conda environment `deepetpicker`	Execution of the ProPicker/DeepETPicker training and inference stack.
Version control	Git	Repository tracking and experiment-code management.

Table A.3: Software, languages and technologies used throughout the project.

Appendix B

Repository Scaffolding and Reproducibility Guidelines

B.1. Current repository scaffolding

The repository was not treated as a passive storage directory during the thesis. It was designed as the scaffolding that makes the experiments reproducible and comparable. Table B.1 summarizes the current structure that matters for the investigation.

Component	Main role	Practical guideline
`paths.py`	Central path registry	Update this file first when moving the project to a new machine. All experiment scripts depend on it.
`config.py`	Shared experimental constants	Located under `experiments/`. Keep common splits and hyperparameters here so that EXP2, EXP3 and EXP4 remain directly comparable.
`exp1_empiar10988_ribo/`	Real-data proof of concept	Folder under `experiments/`. Contains the ribosome notebooks and scripts used to validate the end-to-end pipeline on EMPIAR-10988.
`exp2_umusynth_thy/`	Synthetic single-class baseline	Folder under `experiments/`. Defines the stable UMU thyroglobulin preprocessing and fine-tuning workflow.
`exp3_ppicker_limits/`	Data-efficiency study	Folder under `experiments/`. Stores the incremental-training notebooks, prompt-selection helpers and long-run scripts for the single versus multi prompt comparison.
`exp4_ppicker_rotations/`	Prompt-robustness study	Folder under `experiments/`. Contains the quaternion-based selector, the checkpoint-family evaluator and the underperforming-prompt analysis notebooks.
`results/`	Derived artifacts	Prompts, coordinates, checkpoints, inference outputs and analysis tables should be written here rather than into the source folders.
`docs/`	Documentation and reports	The thesis, interim reports and future experiment specifications are maintained here and versioned together with the code.

Component	Main role	Practical guideline
`models/ + tools/`	External dependencies	Store checkpoints in `models/` and keep the cloned ProPicker code under `tools/`. They act as explicit external inputs to the experiments.

Table B.1: Repository scaffolding used by the thesis workflow.

B.2. Experiment template and naming conventions

The project follows a simple scaffold when a new experiment is added:

1. create a folder named `exp<N>_<dataset>_<goal>` inside `experiments/`;
2. add at least one notebook for preprocessing, inspection or analysis;
3. add a `scripts/` subfolder for long runs;
4. register the new output directories in `paths.py`;
5. reuse `experiments/config.py` whenever the split or hyperparameters should remain comparable to earlier work.

The naming of generated files also follows a practical convention. Fixed prompts are stored as `fixed_prompts_*.json`, experimental prompt variants as files such as `single_instance_prompt.json` or `multi_instance_prompt_n10.json`, the preferred checkpoint is copied as `best_model.ckpt`, and inference outputs preserve the toolchain folder names such as `PredictedLabels` and `Coords_All`. This regularity matters because later notebooks assume those names when aggregating results.

B.3. Operational guidelines

The repository contains not only code, but also workflow rules that turned out to be necessary for reliable experimentation:

- Run the preprocessing notebook before launching the long training script. The scripts assume that prompt files, converted coordinates and configuration templates already exist.
- Launch fine-tuning and inference from the ProPicker internal working directory expected by the DeepETPicker helper scripts. This avoids silent path-resolution failures.
- Keep the validation split fixed once a benchmark is established. In practice, the 20/5 UMU split defined in EXP2 must remain unchanged in EXP3 and EXP4.
- Store derived artifacts in `results/` and keep source code in `experiments/`. Mixing both makes later auditing unnecessarily difficult.
- Export localization maps to MRC whenever volumetric interpretation matters. ParaView then becomes part of the validation workflow, not a post hoc visualization tool.

- Update the written report in `docs/` whenever a new experiment changes the state of the investigation. This prevents the code and the thesis narrative from diverging.

B.4. Minimal reproducibility checklist

For a reader who wants to rerun or extend the present investigation, the minimum checklist is the following:

1. configure `paths.py` for the local machine;
2. place the ProPicker and TomoTwin checkpoints in `models/`;
3. clone ProPicker into `tools/ProPicker/`;
4. install the Python environment used by DeepETPicker and ProPicker;
5. populate `data/` with the EMPIAR or UMU datasets;
6. execute the notebook that generates prompts and coordinates for the target experiment;
7. run the training or inference script corresponding to that experiment;
8. verify scalar outputs and, when relevant, inspect the exported MRC maps in ParaView.

This checklist is intentionally minimal. Its role is not to replace the experiment notebooks, but to make the repository intelligible as a research instrument and to document the scaffolding assumptions on which the thesis results depend.

Appendix C

Condensed Algorithm Listings

The notebooks used in the experiments contain a large amount of supporting code for plotting, file management and interactive inspection. For the appendix, the most useful compromise is to keep only the core algorithmic fragments that define how prompts are scored, selected, aggregated and evaluated. The listings below are therefore condensed from the real experiment code rather than copied verbatim from entire notebooks. Their purpose is to preserve the research logic in a form that is short enough to audit.

C.1. EXP3 prompt scoring and single-prompt selection

The core prompt-selection logic used in `experiments/exp3_ppicker_limits/exp3_ppicker_limits.ipynb` combines three physically motivated features: a mid-frequency versus high-frequency ratio, a center-versus-border energy ratio and a penalty for global offset. In the full notebook, helper functions perform robust normalization and optional embedding-consensus refinement; Listing C.1 keeps only the essential decision rule.

```
def compute_quality_score(subtomo, border=4, centre_frac=0.45,
                          k_low=0.08, k_high=0.30):
    vol = robust_norm(to_numpy(subtomo))
    freq_ratio = freq_structure_ratio(vol, k_low=k_low, k_high=k_high)
    centre_ratio = center_border_ratio(
        vol, border=border, centre_frac=centre_frac
    )
    dc_penalty = abs(vol.mean()) / (vol.std() + 1e-6)

    score = np.log(freq_ratio) + 0.9 * np.log(centre_ratio) - 0.6 * dc_penalty
    return score, {
        "freq_ratio": freq_ratio,
        "centre_ratio": centre_ratio,
        "dc_penalty": dc_penalty,
    }


def select_best_prompt(subtomos, prompt_encoder=None, top_k=8):
    base_scores = np.array([compute_quality_score(v)[0] for v in subtomos])
    candidate_idx = np.argsort(base_scores)[::-1][:top_k]
    final_scores = base_scores.copy()

    if prompt_encoder is not None and len(candidate_idx) >= 2:
```

```
        E = np.stack([prompt_encoder(subtomos[i]).reshape(-1)
                      for i in candidate_idx])
        S = cosine_similarity_matrix(E)
        consensus = (S.sum(axis=1) - 1.0) / (S.shape[0] - 1)
        for j, i in enumerate(candidate_idx):
            final_scores[i] += 1.2 * consensus[j]

    best_idx = int(np.argmax(final_scores))
    return best_idx, subtomos[best_idx]
```

Listing C.1: Condensed EXP3 single-prompt selector based on subtomogram quality.

This selector reflects an important methodological shift in the thesis: once detector performance becomes competitive, prompt choice itself becomes part of model selection rather than a manual convenience step.

C.2. EXP3 multi-prompt aggregation

The second important idea introduced in `experiments/exp3_ppicker_limits/exp3_ppicker_limits_inference.ipynb` is that the operational prompt does not have to be a single subtomogram embedding. In the experiment, several prompt embeddings are computed from selected particles and their arithmetic mean is used as the «multi-prompt» representation. Listing C.2 shows the compact form of that logic.

```
def build_prompt_embeddings(subtomos, prompt_encoder, n_instances=10):
    embeddings = torch.stack([
        prompt_encoder(subtomo).reshape(-1)
        for subtomo in subtomos[:n_instances]
    ])

    single_prompt = embeddings[0]
    multi_prompt = embeddings.mean(dim=0)

    return {
        "single_prompt": single_prompt,
        "multi_prompt": multi_prompt,
        "all_embeddings": embeddings,
    }
```

Listing C.2: Condensed EXP3 multi-prompt aggregation. In the notebook, the input subtomograms are already pre-selected by the prompt-scoring stage.

In the actual experiment, this averaged embedding produced the best observed operating mode because it reduced false positives more strongly than it reduced recall.

C.3. EXP4 quality-filtered rotation-diverse prompt selection

The final version of `experiments/exp4_ppicker_rotations/exp4_large_prompt_population_inference.ipynb` extends prompt selection by adding quaternion distances and a symmetry-aware interpretation for thyroglobulin. The selector first filters out unsafe border candidates, then keeps only high-quality subtomograms, and finally builds a prompt population that is as diverse as possible in orientation space while still preferring better-quality candidates. Listing C.3 gives the condensed form.

```python
def quaternion_angular_distance_c2(q1, q2, c2_rotation):
    r1 = Rotation.from_quat(normalize_quaternion(q1))
    r2 = Rotation.from_quat(normalize_quaternion(q2))
    return min(
        np.degrees((r1.inv() * r2).magnitude()),
        np.degrees((r1.inv() * (r2 * c2_rotation)).magnitude()),
    )


def select_diverse_rotations_with_quality(
    df, tomo_cache, n_samples, min_angular_dist=30.0,
    quality_percentile=50.0, max_per_tomo=2
):
    df = filter_border_safe_candidates(df, tomo_cache, min_border_margin=50)
    df["quality_score"] = [
        compute_quality_score(extract_subtomo(row, tomo_cache))[0]
        for _, row in df.iterrows()
    ]

    threshold = np.nanpercentile(df["quality_score"], quality_percentile)
    df = df[df["quality_score"] >= threshold].copy()

    quats = df[["Q1", "Q2", "Q3", "Q4"]].to_numpy()
    scores = df["quality_score"].to_numpy()
    tomo_names = df["tomo_name"].to_numpy()

    selected = [int(np.argmax(scores))]
    tomo_counts = {tomo_names[selected[0]]: 1}

    while len(selected) < min(n_samples, len(df)):
        best_idx, best_min_dist, best_quality = -1, -1.0, -np.inf
        for i in range(len(df)):
            if i in selected:
                continue
            if tomo_counts.get(tomo_names[i], 0) >= max_per_tomo:
                continue

            min_dist = min(
                quaternion_angular_distance_c2(
                    quats[i], quats[j], THYROGLOBULIN_C2_ROTATION
                )
                for j in selected
            )

            if min_dist >= min_angular_dist and (
                min_dist > best_min_dist or
                (min_dist == best_min_dist and scores[i] > best_quality)
            ):
                best_idx, best_min_dist, best_quality = i, min_dist, scores[i]

        if best_idx == -1:
            break

        selected.append(best_idx)
        tomo_counts[tomo_names[best_idx]] = (
```

```
            tomo_counts.get(tomo_names[best_idx], 0) + 1
        )

    return df.iloc[selected].copy()
```

Listing C.3: Condensed EXP4 selector for quality-filtered and rotation-diverse prompts.

The key research idea here is not only the use of quaternions, but the fact that the selector is symmetry-aware and quality-aware at the same time. This is precisely what made it possible to separate prompt quality effects from naive «bad rotation» explanations in the final analysis.

C.4. Fixed evaluation rule used in the synthetic experiments

Both `exp3_ppicker_limits_inference.ipynb` and `exp4_large_prompt_population_inference.ipynb` keep a fixed center-matching rule between predicted coordinates and ground truth. Listing C.4 shows the compact implementation. The matching is greedy and one-to-one: once a ground-truth point has been matched, it cannot be reused by another prediction.

```
def compute_metrics(pred_coords, gt_coords, distance_threshold=18.5):
    if len(pred_coords) == 0:
        return {"precision": 0.0, "recall": 0.0, "f1": 0.0,
                "tp": 0, "fp": 0, "fn": len(gt_coords)}
    if len(gt_coords) == 0:
        return {"precision": 0.0, "recall": 0.0, "f1": 0.0,
                "tp": 0, "fp": len(pred_coords), "fn": 0}

    distances = cdist(pred_coords, gt_coords)
    matched_gt = set()
    tp = 0

    for i in range(len(pred_coords)):
        best_j, best_d = -1, np.inf
        for j in range(len(gt_coords)):
            if j not in matched_gt and distances[i, j] < best_d:
                best_j, best_d = j, distances[i, j]

        if best_j != -1 and best_d <= distance_threshold:
            matched_gt.add(best_j)
            tp += 1

    fp = len(pred_coords) - tp
    fn = len(gt_coords) - tp
    precision = tp / (tp + fp) if tp + fp else 0.0
    recall = tp / (tp + fn) if tp + fn else 0.0
    f1 = (2 * precision * recall / (precision + recall)
          if precision + recall else 0.0)

    return {"precision": precision, "recall": recall, "f1": f1,
            "tp": tp, "fp": fp, "fn": fn}
```

Listing C.4: Condensed greedy one-to-one coordinate matching used to compute precision, recall and F_1.

Preserving this rule across the synthetic experiments matters scientifically: it keeps the benchmark constant while the prompt strategy, checkpoint family, or number of training tomograms changes.